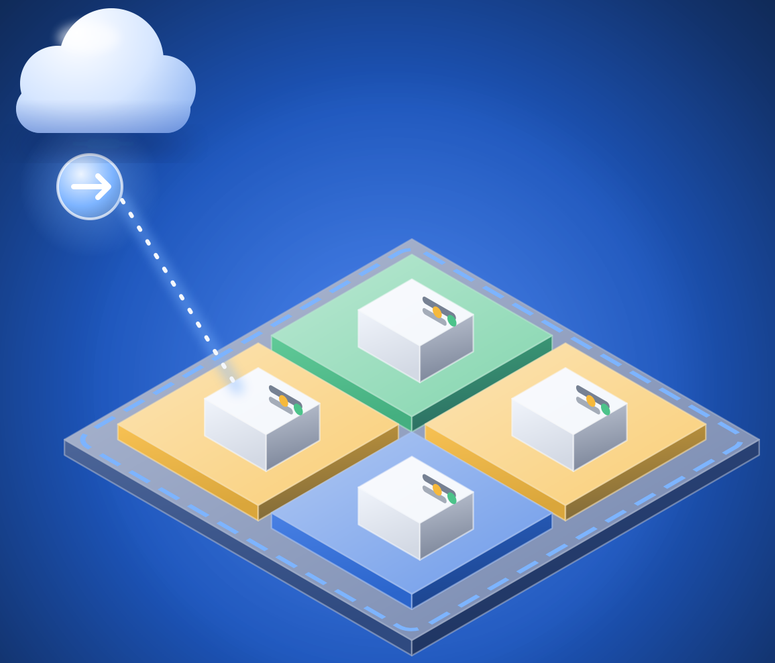
Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 6

Networking in the Cloud

IP addresses, CIDR and subnets, virtual networks, and Amazon VPC.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Recap

LAST SESSION

Storage

A common cloud service: remote services that store our data.

LAST SESSION

Object, block and file storage

The three types of cloud storage, and when to use each.

Learning objectives

1. Introduction to networking
2. Introduction to Amazon VPC
3. Using VPC
4. Introduction to CDN

THIS WEEK

Network

Describe common cloud service models: network. We discuss the common concepts of networking in the cloud.

PART 1

Introduction to networking

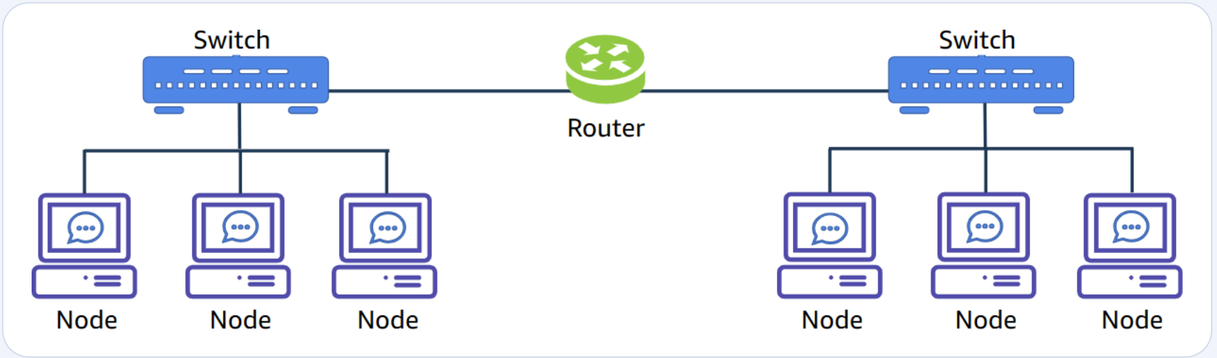
Nodes, links and protocols, then IP addresses, CIDR blocks and subnets.

Week 6 · 04

What is networking?

A network is a set of *interconnected devices* sharing resources and data.

The devices communicate using communication protocols over a physical or wireless connection.

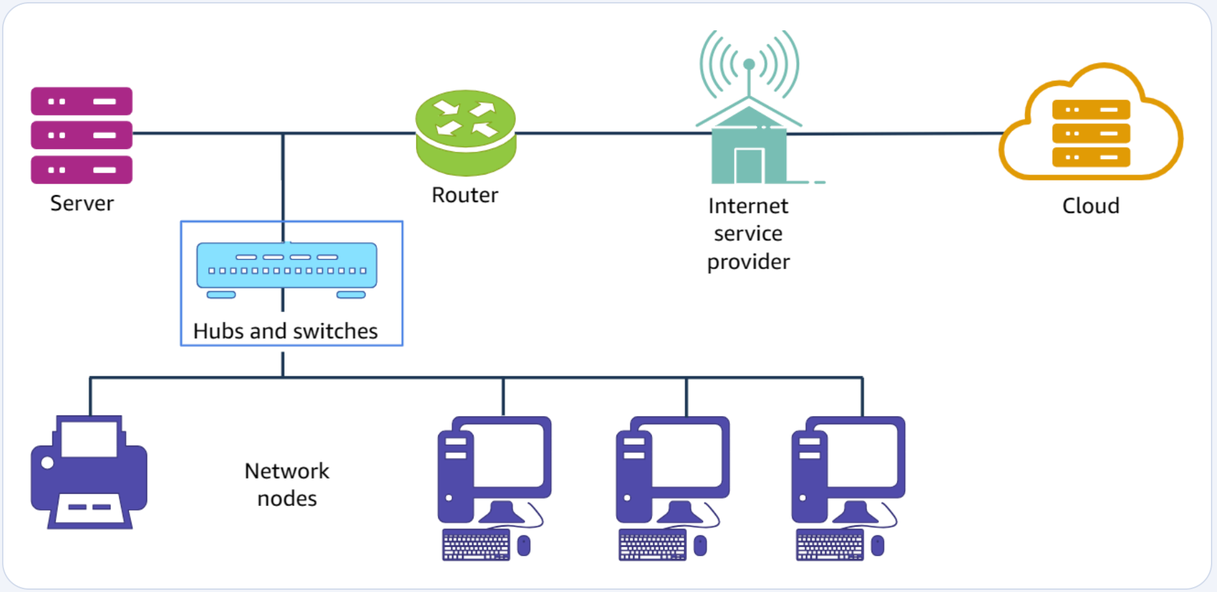


Nodes and links

Networks are integral to our lives, at home, in the office and in public places.

The devices can be computers, switches, routers, printers and servers. We call them nodes, and each node needs an identifier, often an IP address.

A link is the transmission media connecting two nodes: cable wire, fiber cable or wireless.



Communication protocols

Communication protocols define the rules for exchanging information over the network. TCP/IP are the most important.

TCP/IP layer	Protocols
Application layer	HTTP, FTP, DNS, SNMP, Telnet, SMTP, POP, IMAP
Transport layer	TCP, UDP
Internet layer	IP (IPv4, IPv6), ICMP, ARP
Physical layer	Ethernet, PPP, ATM

TCP/IP is a concise version of the OSI model: four layers, compared to seven in the OSI model.

LAN vs WAN

Networks come in different sizes and shapes and are used differently.

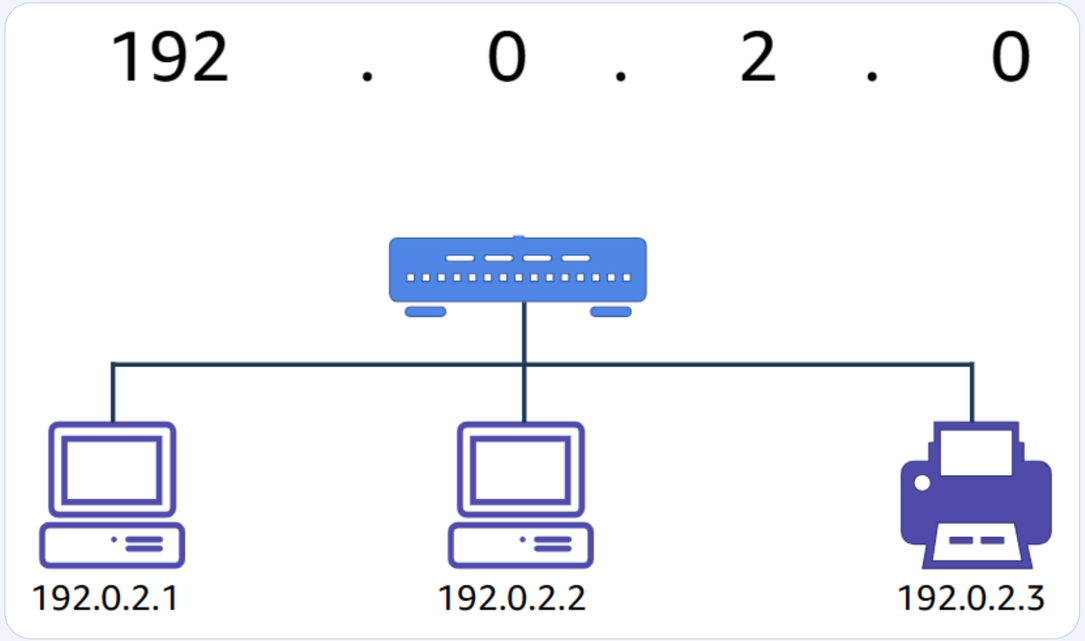
	LAN (local area network)	WAN (wide area network)
Ownership	It is privately owned.	It can be managed by multiple organizations, since locations are dispersed.
Speed	Faster speed (10 GB or more).	Slower speed (about 1 GB).
Congestion	It is less congested.	It is more congested.
Management	It can be managed in-house.	It requires a third party to configure the setup, and can be costly.

IP address

An IP address is a *numerical label* in decimal format.

Each node in a network has a unique IP address that identifies it.

Here three nodes on one switch are 192.0.2.1, 192.0.2.2 and 192.0.2.3.



IP address structure

192 . 0 . 2 . 26

Bit number	1	2	3	4	5	6	7	8
Bit value placeholder	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Bit value selected	1	1	0	0	0	0	0	0

128 + 64 = 192

192 . 0 . 2 . 26

Bit number	1	2	3	4	5	6	7	8
Bit value placeholder	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Bit value selected	0	0	0	0	0	0	0	0

No bit values selected = 0

Each of the four sections is 8 bits. The selected bit values are added together to give the decimal number.

IP address structure: the last two sections

192 . 0 . 2 . 26

Bit number	1	2	3	4	5	6	7	8
Bit value placeholder	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Bit value selected	0	0	0	0	0	0	1	0

Only 2 is selected = 2

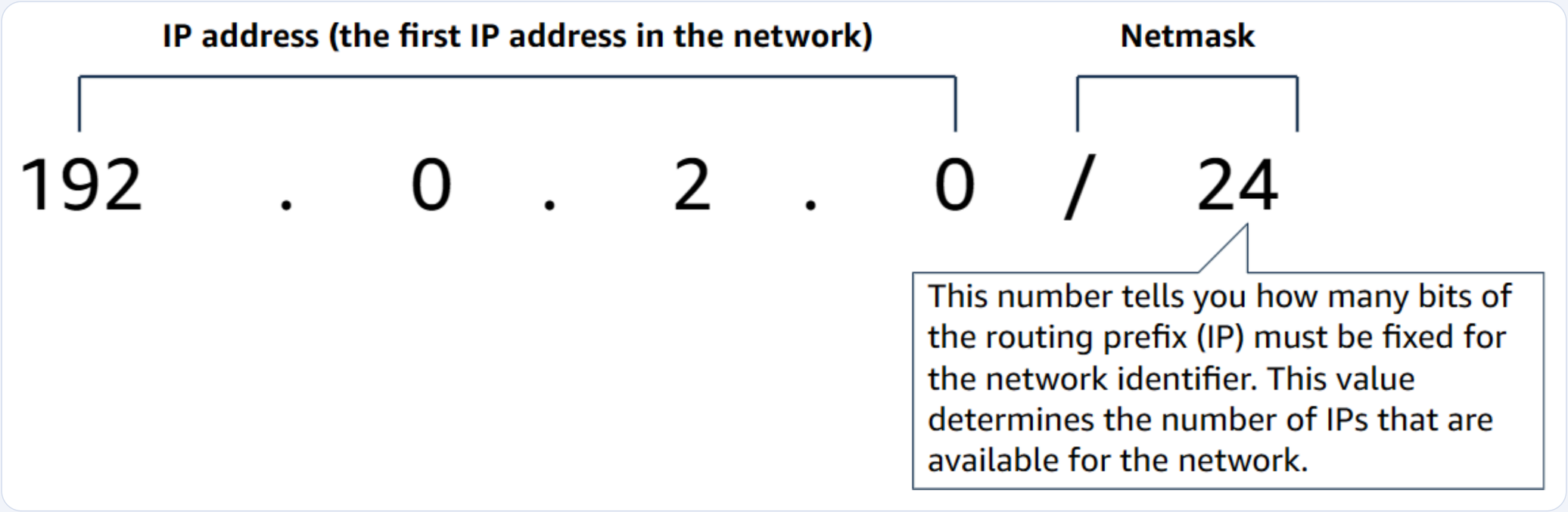
192 . 0 . 2 . 26

Bit number	1	2	3	4	5	6	7	8
Bit value placeholder	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Bit value selected	0	0	0	1	1	0	1	0

16 + 8 + 2 = 26

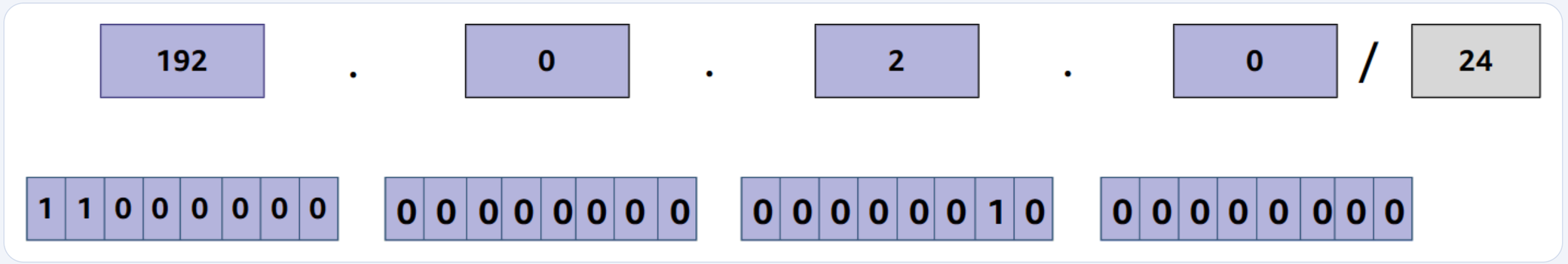
Four sections of 8 bits make a 32-bit address: 192.0.2.26.

Classless Inter-Domain Routing (CIDR)



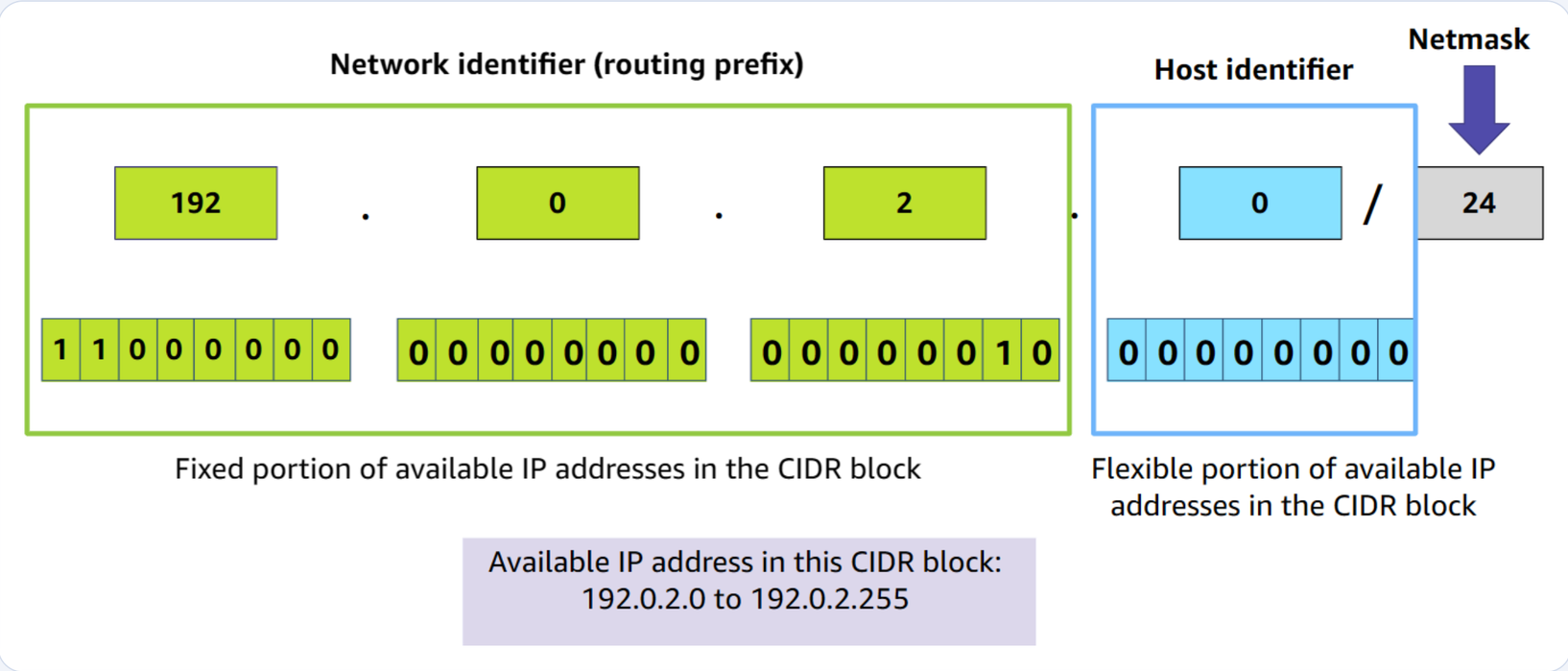
A CIDR block is the first IP address in the network plus a netmask that says how many bits are fixed.

CIDR block example 1: 192.0.2.0/24



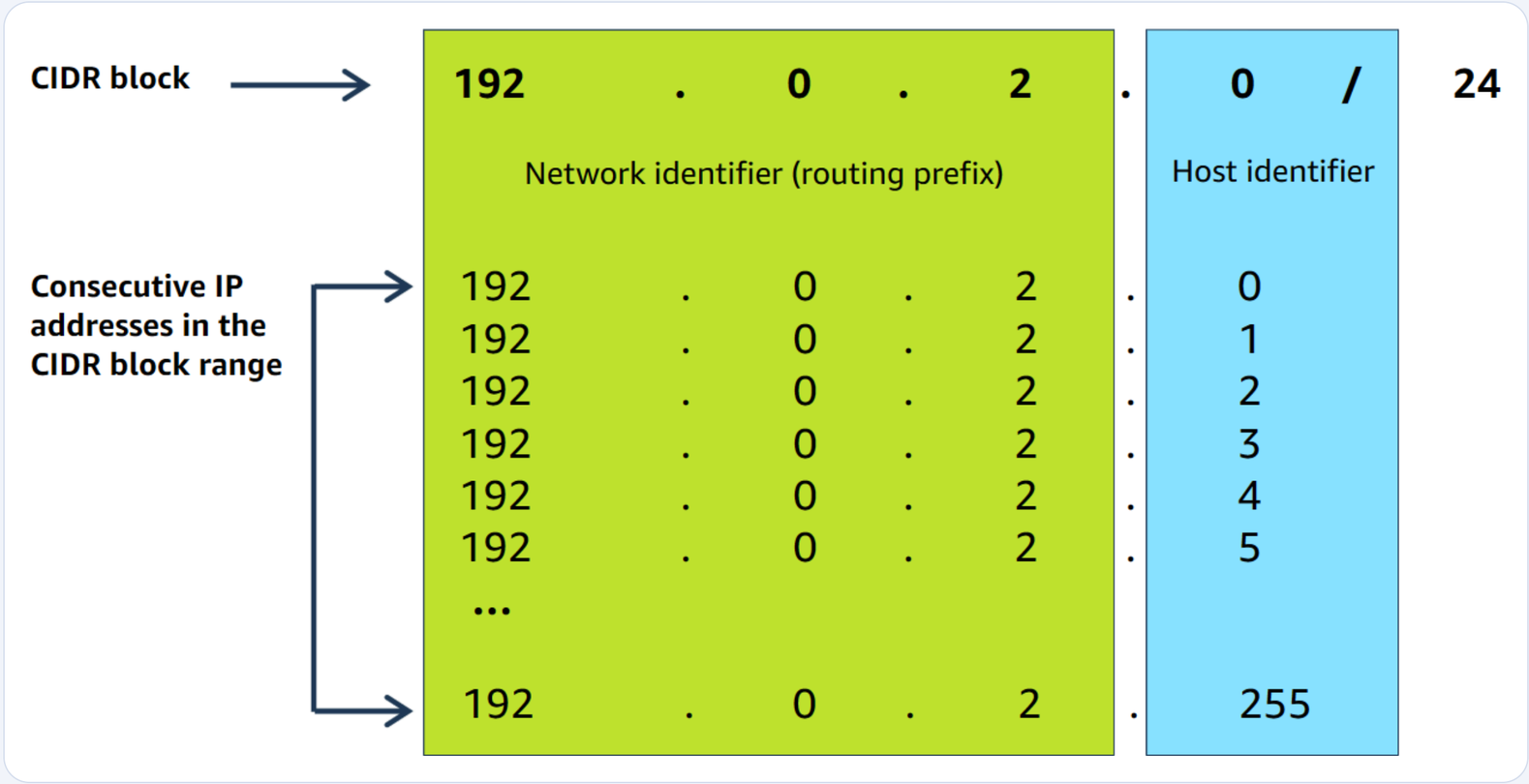
The address is 32 bits: four sections of 8. The /24 says the first 24 bits are fixed.

CIDR block example 1: fixed and flexible

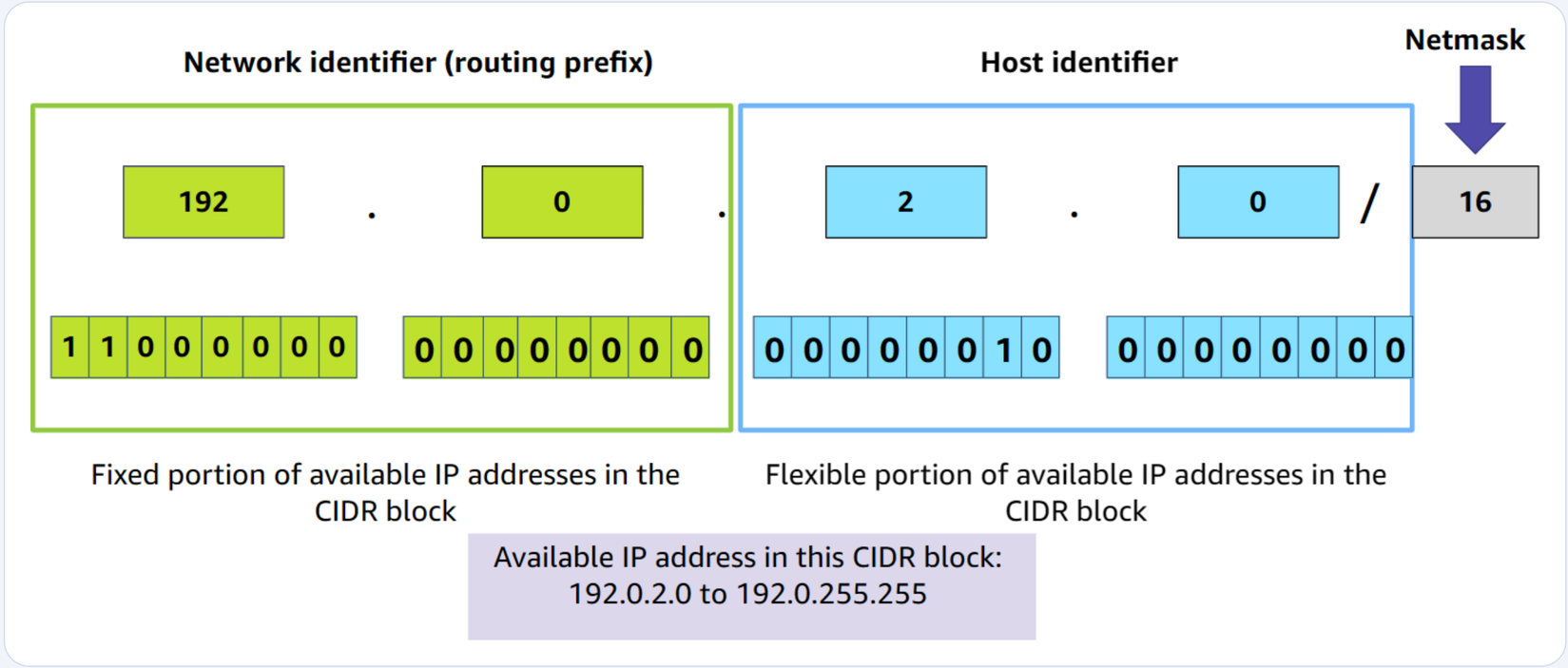


24 fixed bits form the network identifier. The 8 flexible bits give 256 addresses.

CIDR block example 1: the range

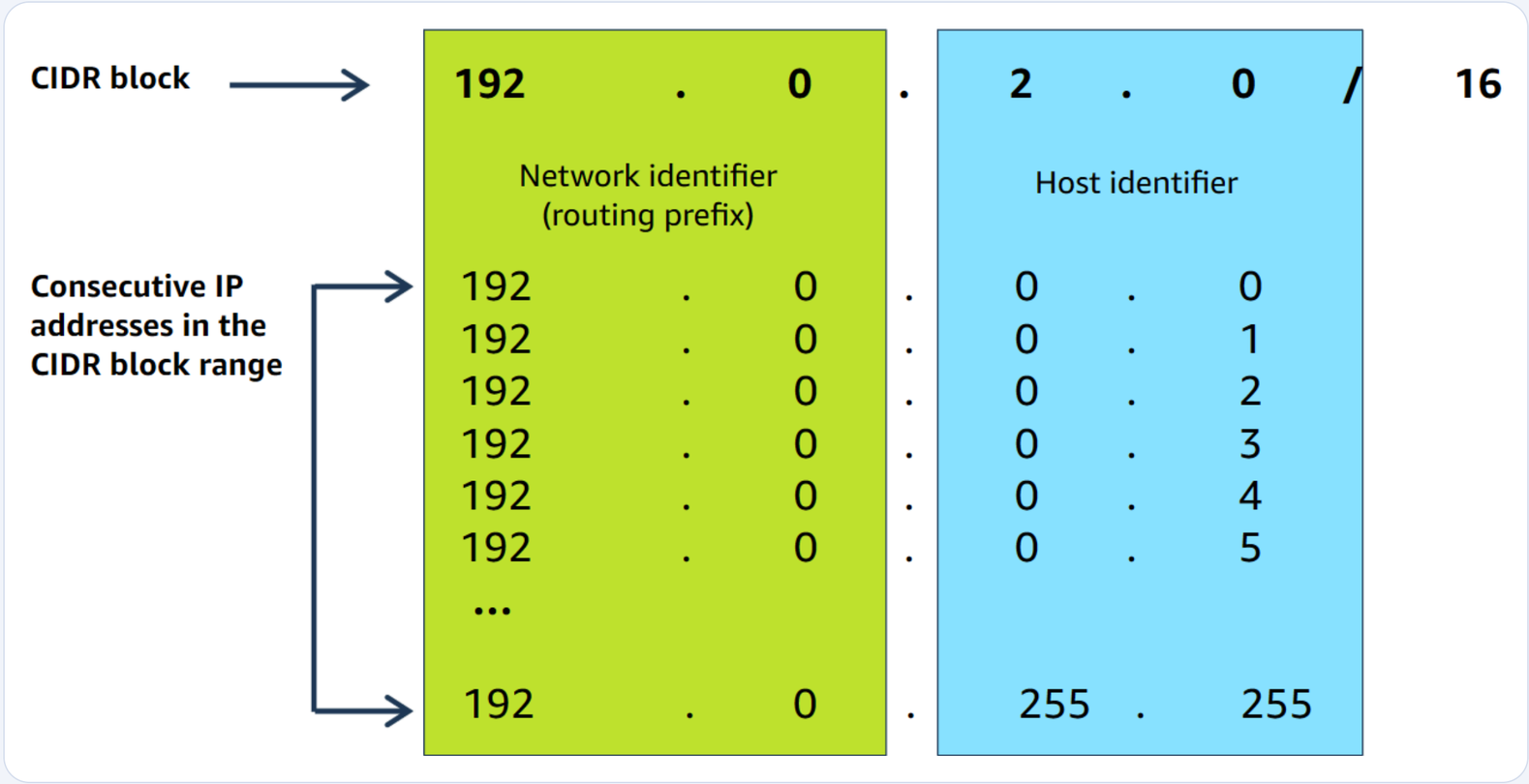


CIDR block example 2: 192.0.0.0/16

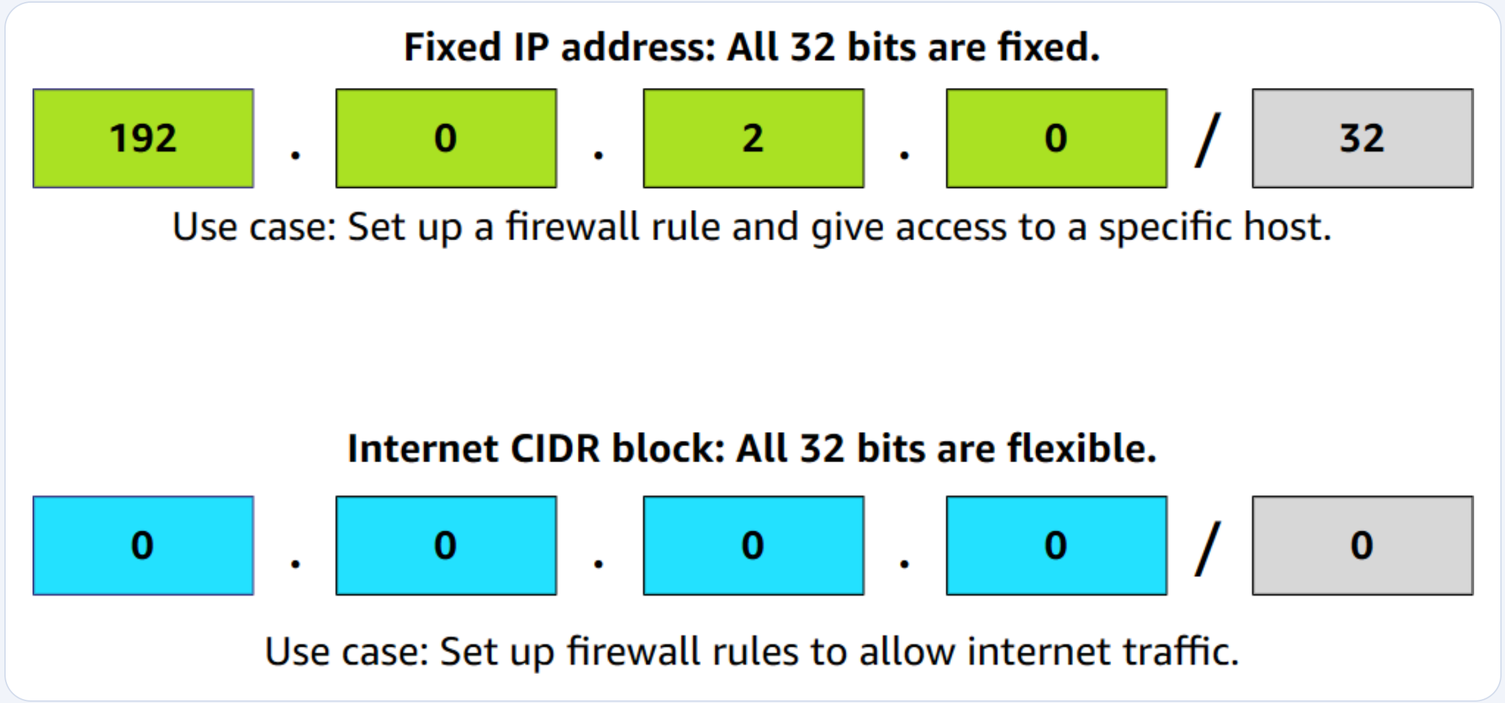


A smaller netmask fixes fewer bits, so the block holds more addresses: 65,536 with /16.

CIDR block example 2: the range

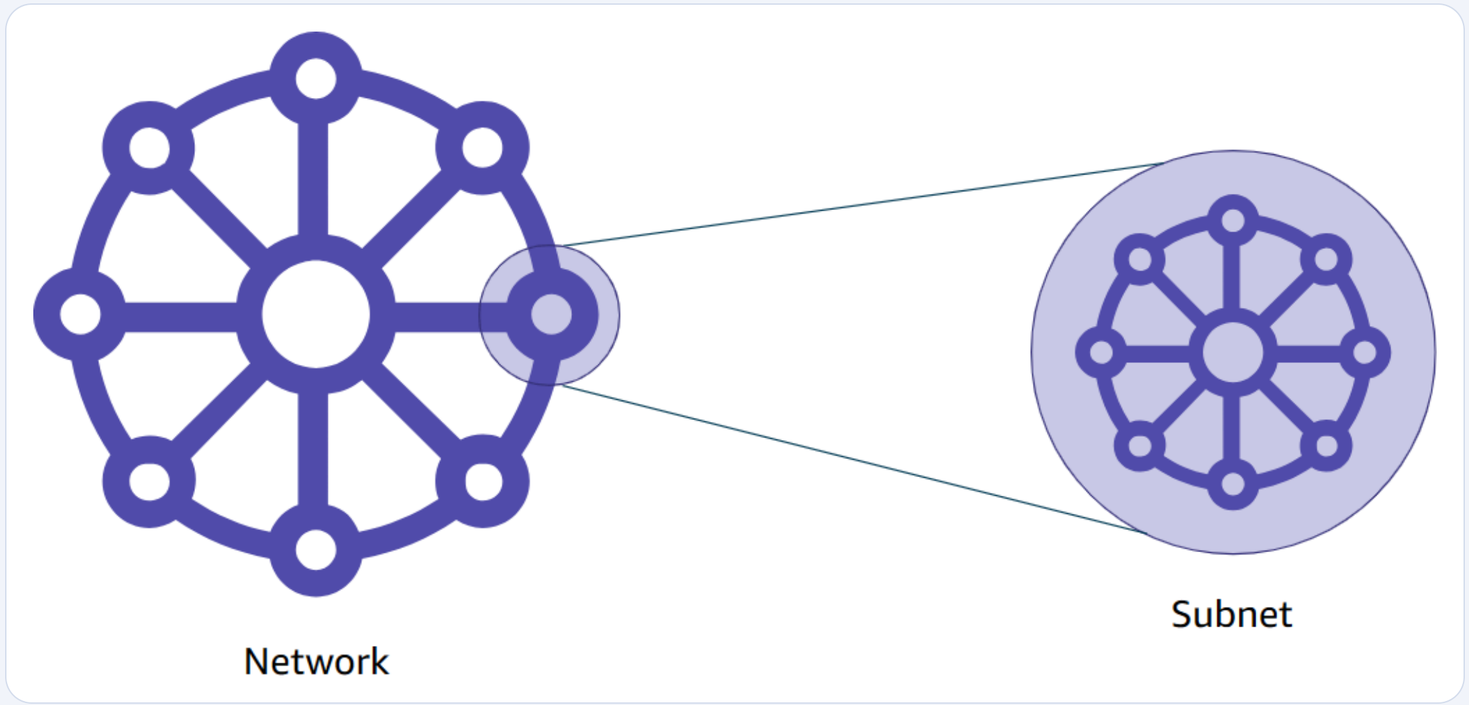


CIDR special cases



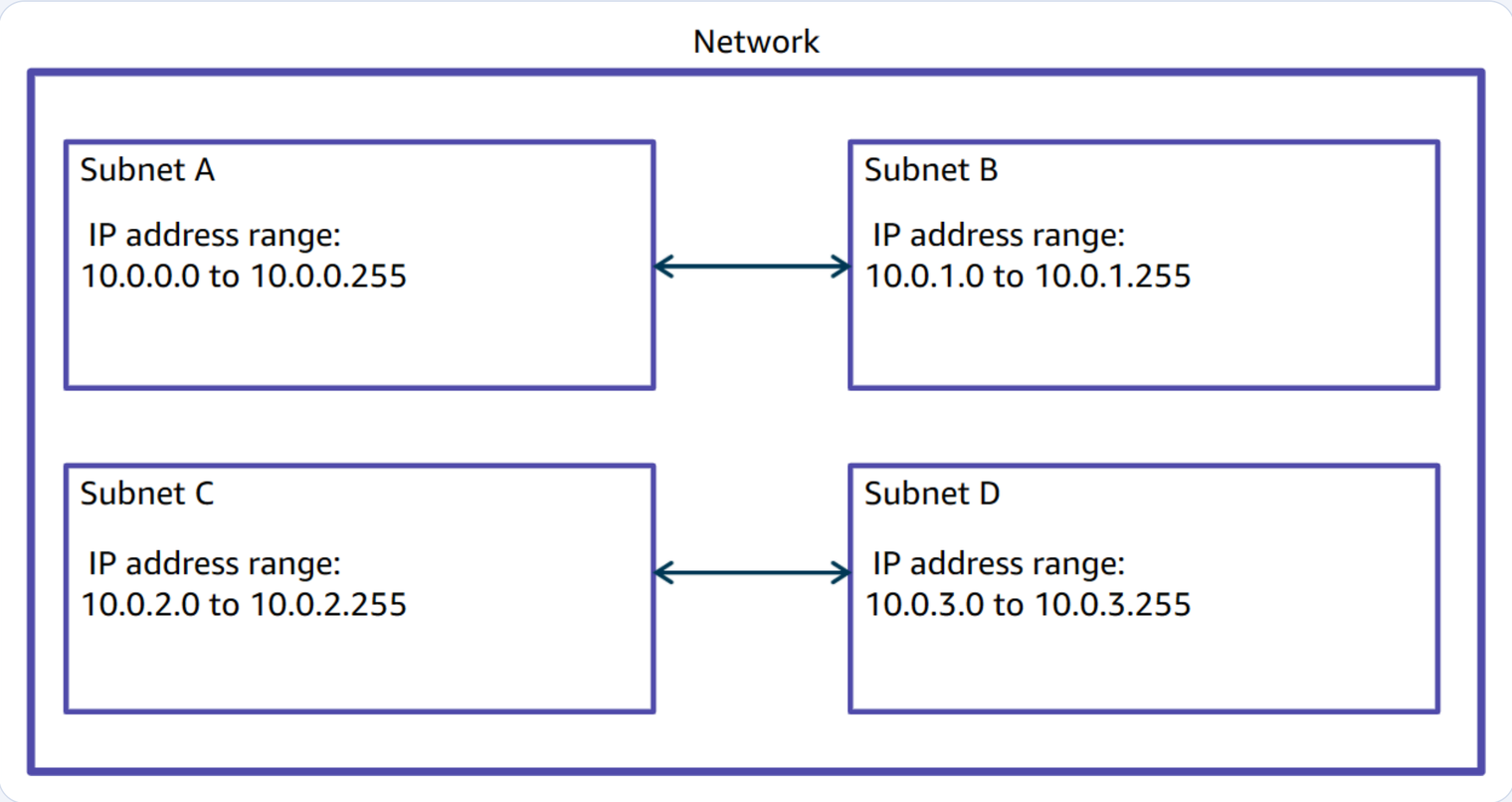
/32 is exactly one host. 0.0.0.0/0 is every address on the internet.

Subnets

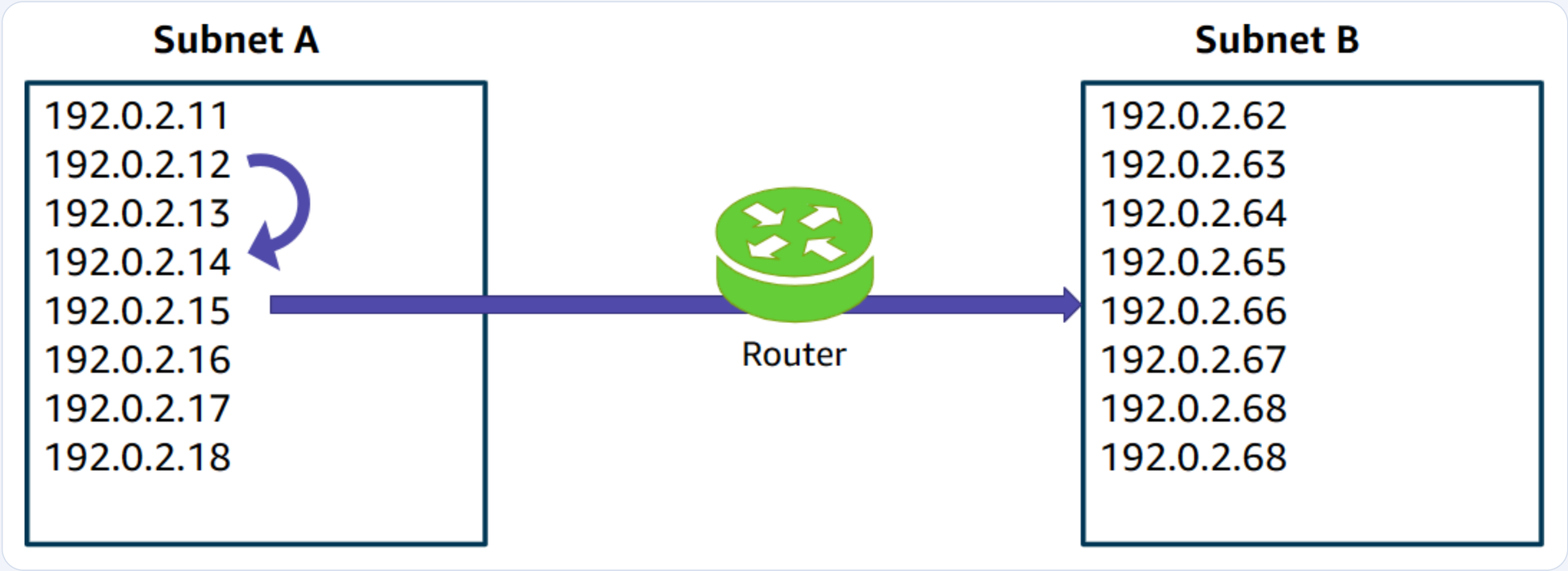


A subnet is a smaller network inside a network.

Subnets: one range each

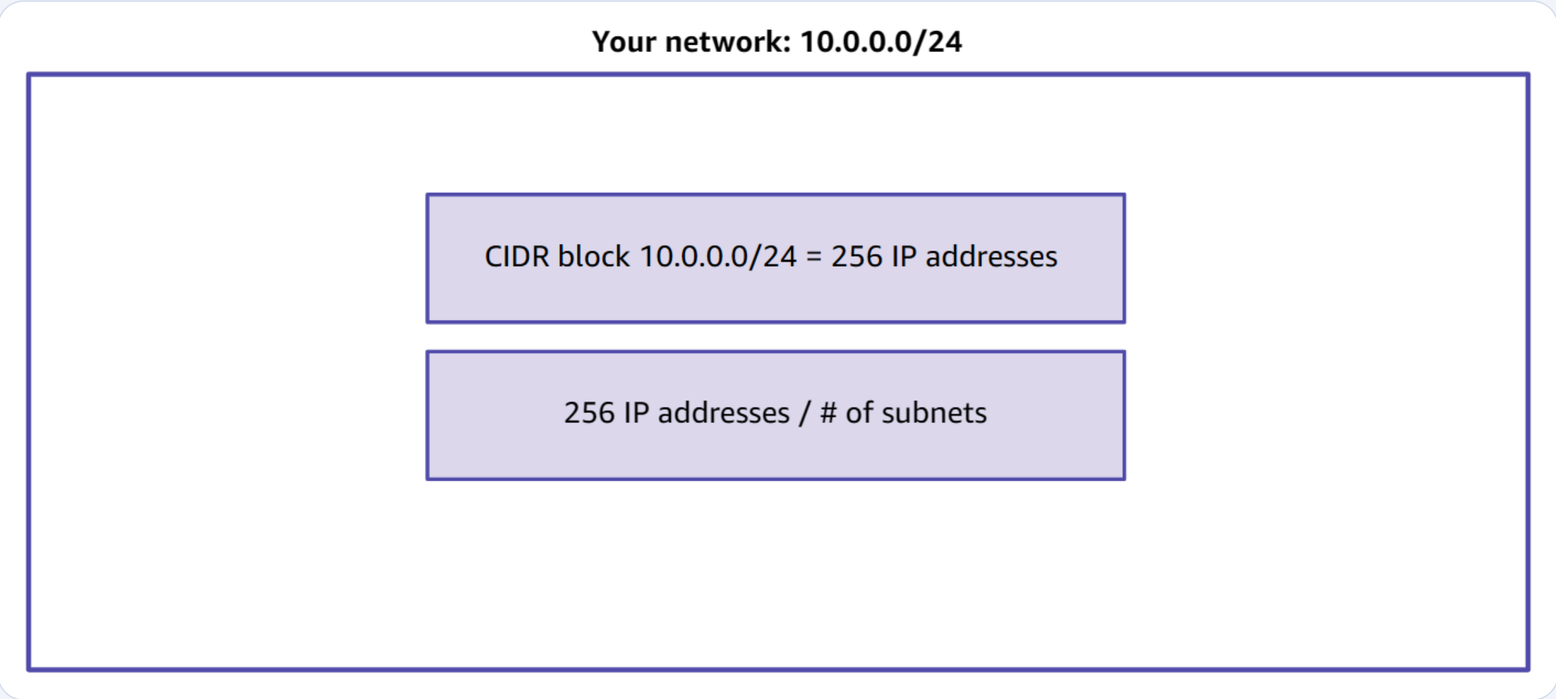


Subnets: traffic between them



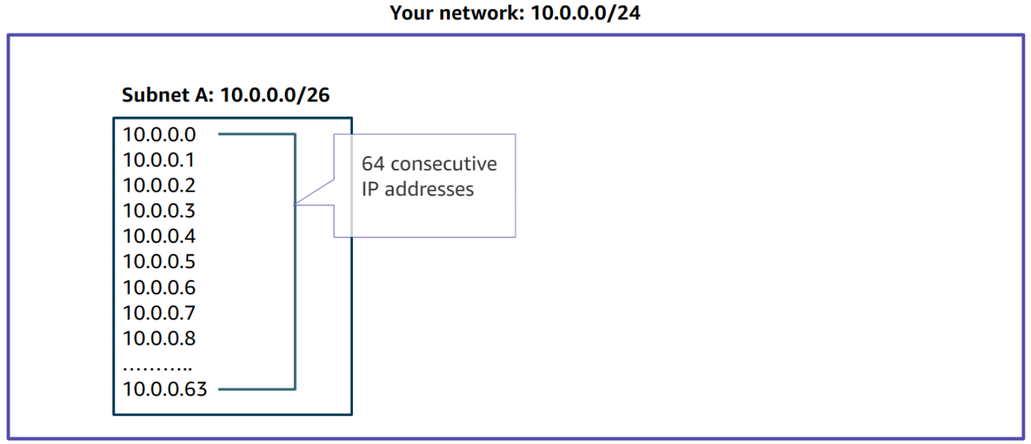
Devices in one subnet reach each other directly. Traffic to another subnet goes through a router.

Subnet CIDR blocks

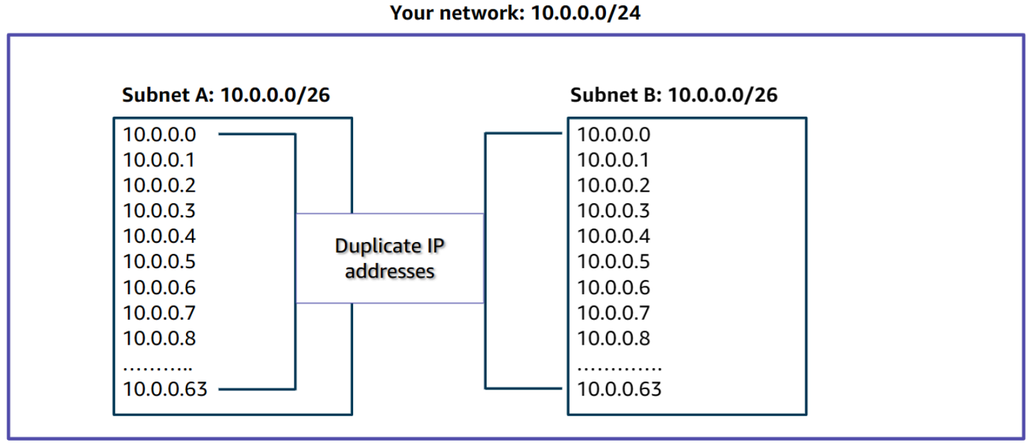


A subnet takes a share of the network’s addresses: 256 IP addresses divided by the number of subnets.

Subnet CIDR blocks: no overlap



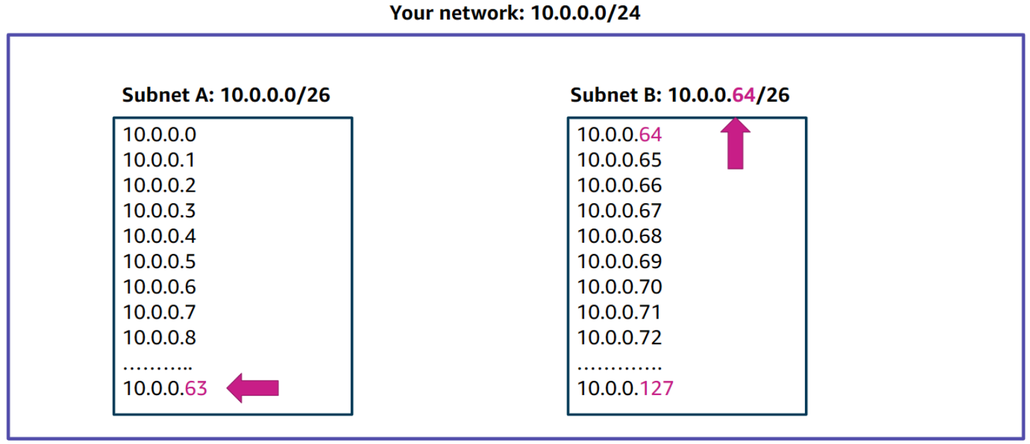
A /26 subnet holds 64 addresses



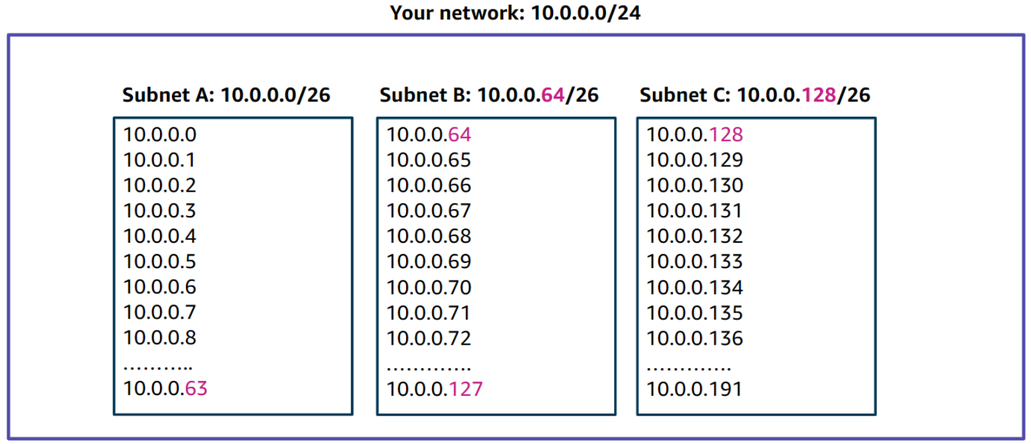
Same block twice: duplicate IP addresses

Two subnets in one network must never share addresses.

Subnet CIDR blocks: the next block



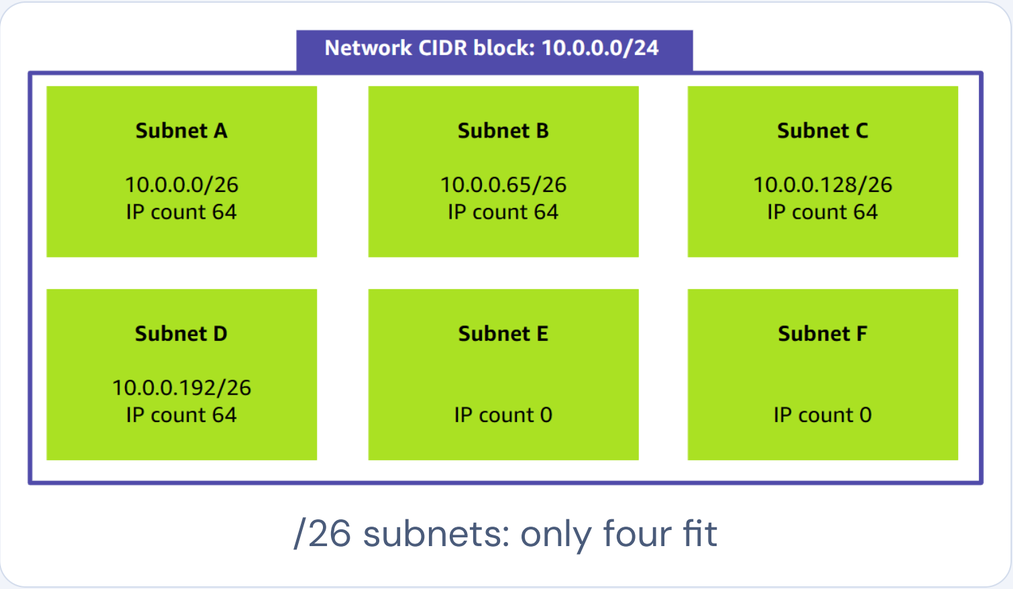
Subnet B starts where A ends



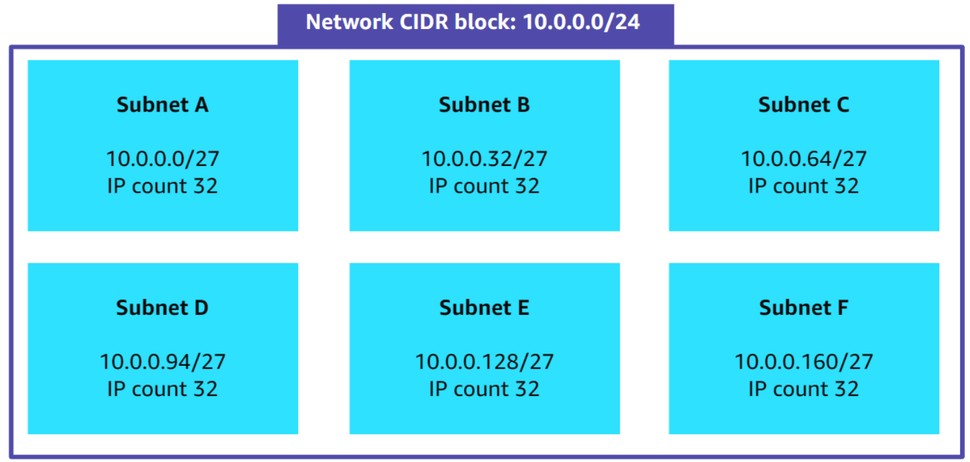
Three subnets, 64 addresses each

Each new subnet starts at the address after the last one of the subnet before it.

Planning ahead



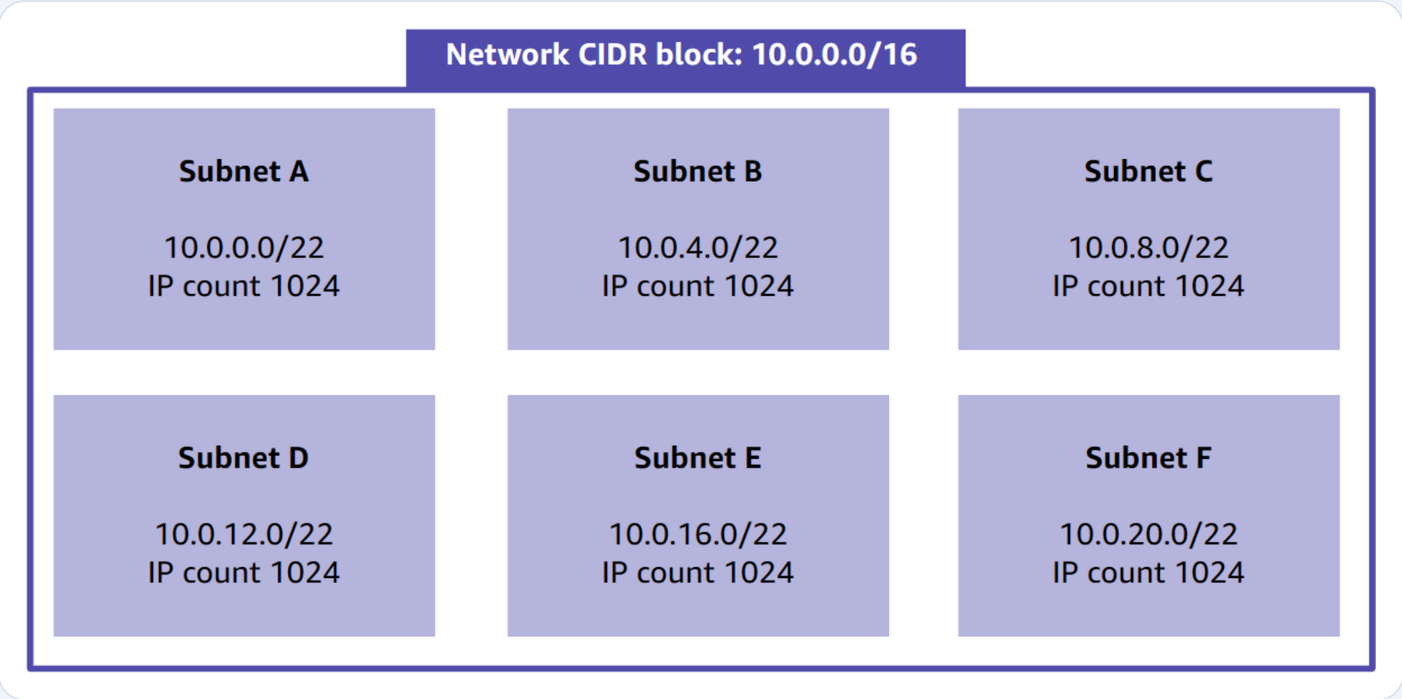
/26 subnets: only four fit



/27 subnets: six fit, 32 each

Decide how many subnets you need before you choose their size.

Planning ahead: a larger network



A /16 network leaves room to grow: six /22 subnets of 1,024 addresses, with space to spare.

Subnet types



Public subnet

Allows internet traffic that is routed through an internet gateway to reach the subnet. A good choice if you have a website that is targeting customers.



Private subnet

Denies traffic routed from the public internet. Use it when your network must reach outside services but outside services must not start a connection to it. Internet access requires a NAT device.

PART 2

Networking in the cloud

The same ideas, delivered by a cloud platform as software.

Week 6 · 28

Cloud network

A cloud network can be considered a WAN where a *cloud-based service* delivers the infrastructure.

On demand

Organizations host resources in a public or private cloud platform and access them on demand. In a hybrid cloud, some resources can still be on-premises.

What is involved

Virtual routers, firewalls, switches and management software are some of the resources involved in networking.

Why customers choose it

It helps them accelerate their time to market, scale the infrastructure on demand and manage the cost of operations.

Software instead of hardware

Cloud networking is achieved mainly via *software* rather than hardware.

Replicated in software

In a cloud computing environment, vendors replicate hardware constructs via software.

A digital representation

We can think of it as a digital representation of network hardware such as routers, switches, firewalls and gateways.

Abstracted

Cloud networking abstracts all the hardware details and uses a software-based approach to set up networks.

Virtual networks

A cloud network is better known as a virtual network. It emulates a physical network, with hardware abstracted via software.

Simpler

It simplifies the connectivity and management of network infrastructure.

The platform's backbone

Traffic is routed on the cloud platform's own backbone rather than traversing the internet, which makes it more secure and faster.

Isolated boundaries

Like traditional networks, virtual networks let us create logically isolated boundaries where we run cloud resources.

Once we are connected to the closest datacenter of a cloud platform, everything after that entry point is on the platform's backbone.

Advantages of virtual networks



Lower cost



Scalability



Productivity



Improved security

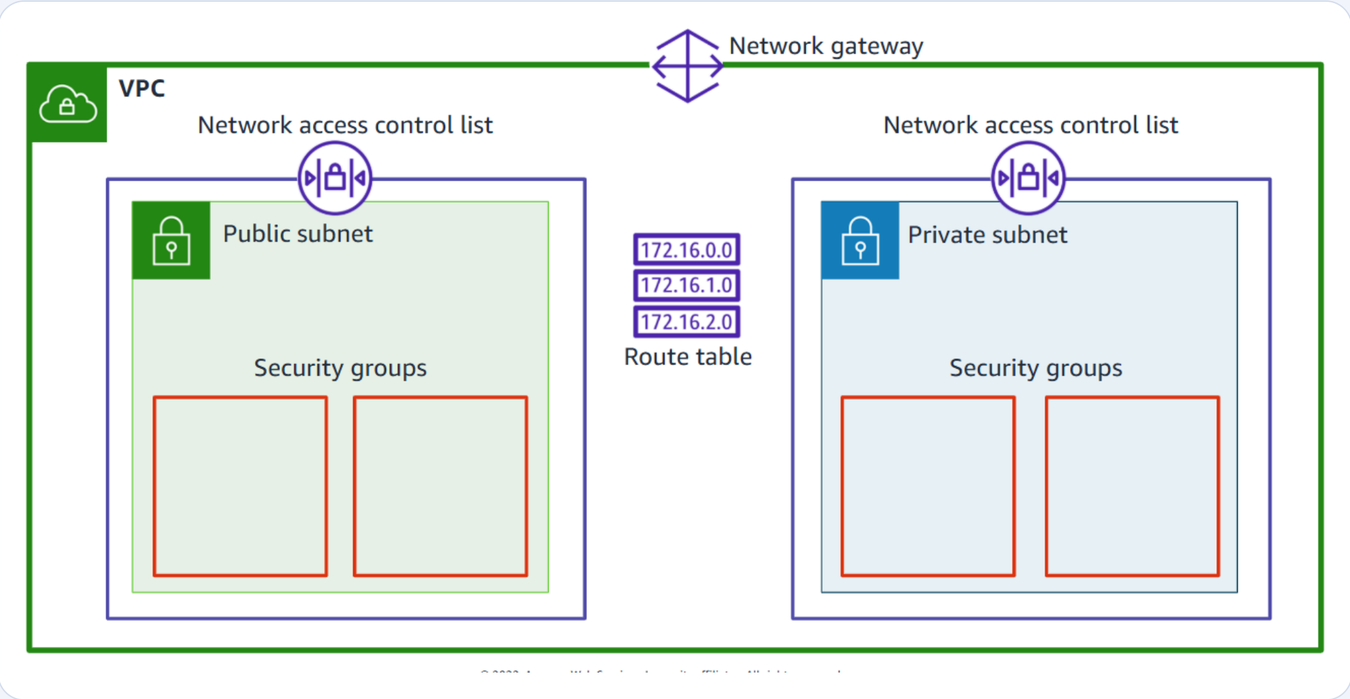
PART 3

Amazon VPC

The virtual network on AWS: CIDR blocks, subnets, gateways, route tables and firewalls.

Week 6 · 33

Amazon VPC



A VPC is your own logically isolated network on AWS. These are the parts we meet in this session.

Amazon VPC CIDR blocks

Choosing the right CIDR block size for your VPC.

/16

Largest AWS CIDR block

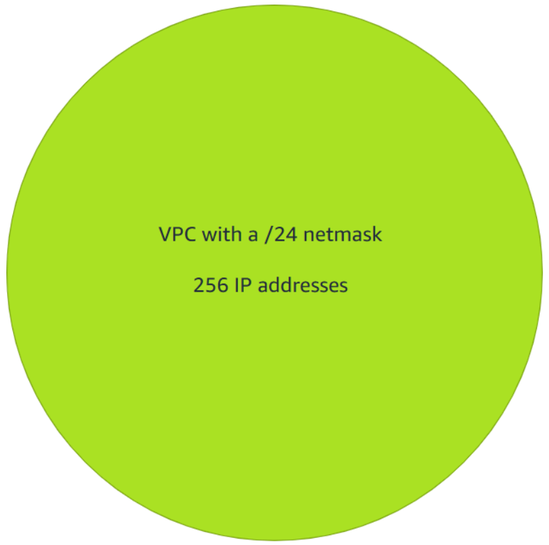
65,536 IP addresses.

/28

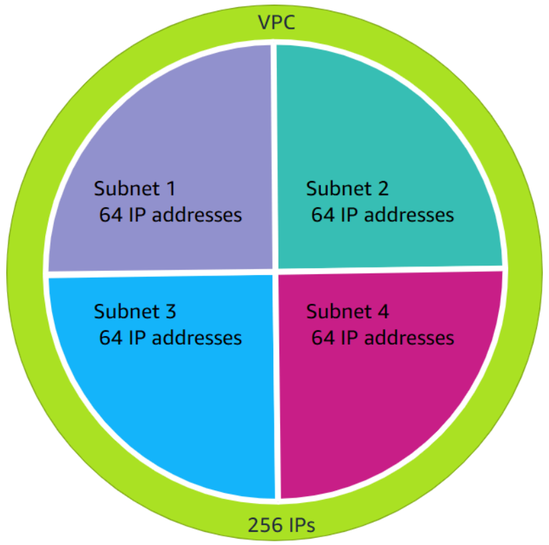
Smallest AWS CIDR block

16 IP addresses.

Amazon VPC CIDR blocks: dividing a VPC

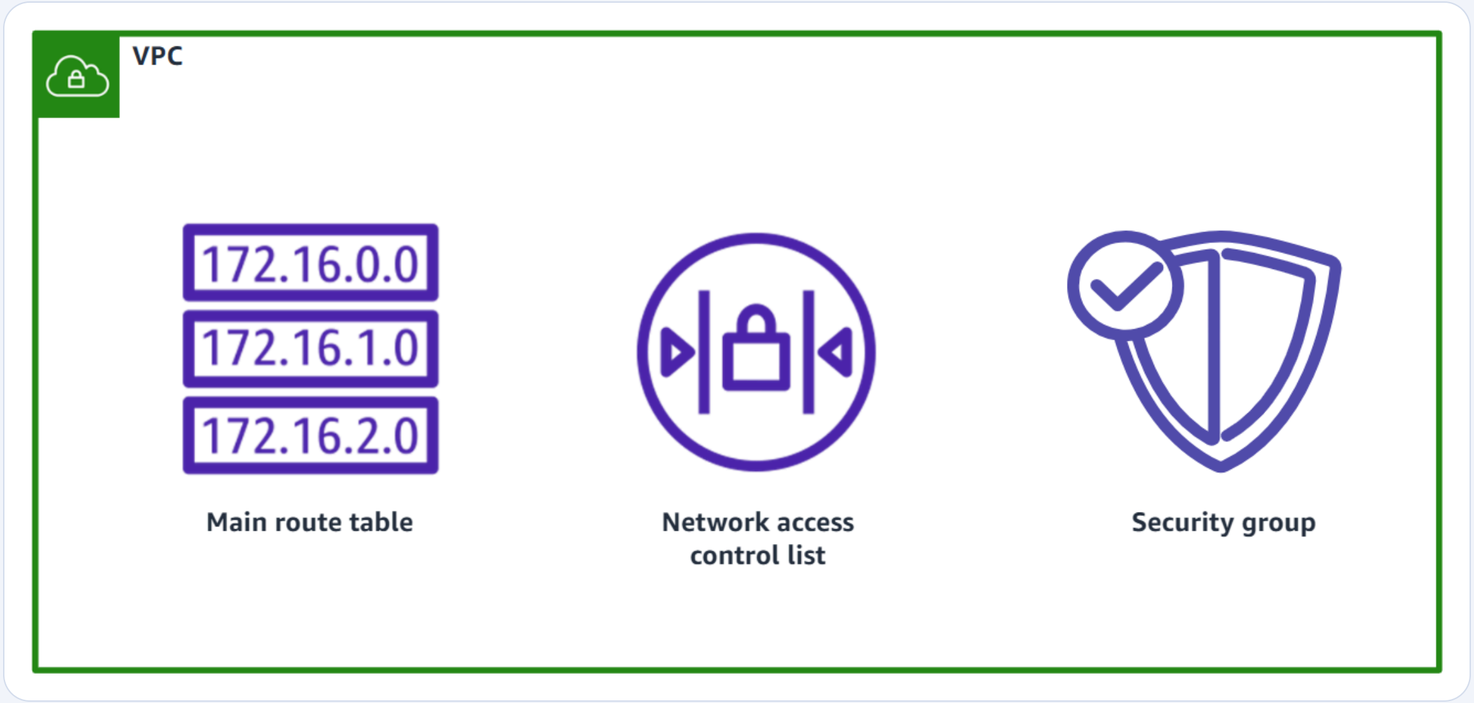


A /24 VPC: 256 IP addresses



Four subnets of 64 IP addresses

Amazon VPC default components



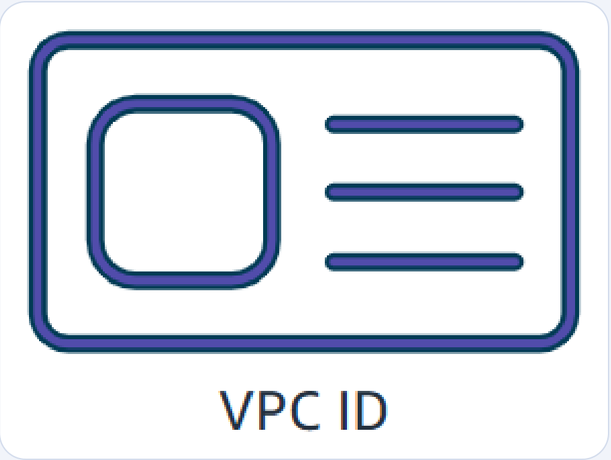
Every VPC is created with a main route table, a network access control list and a security group.

VPC ID

When you create a VPC, it is auto-assigned a VPC ID: a string of random numbers and letters that identifies your VPC.

The ID is required when you associate additional components to your VPC, such as subnets and route tables. VPC IDs cannot be edited.

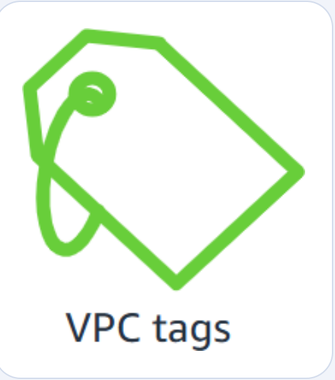
VPC ID example: **vpc-0476e890abedg995f**



VPC tags

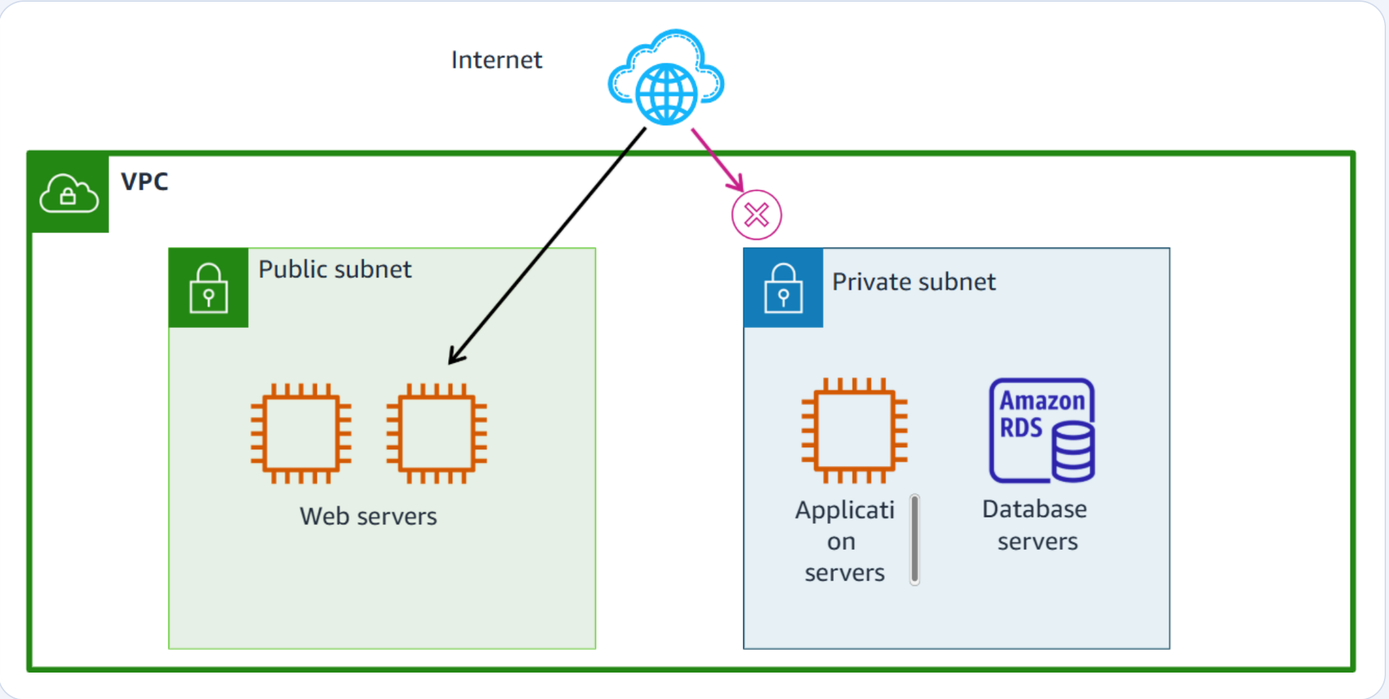
Because the VPC ID can be difficult to remember, you can tag your VPC with a more identifiable name. The tag name appears after the VPC ID in parentheses.

You can tag every component of your VPC, including subnets, route tables and network ACLs. Tagging is recommended, to be sure you connect the correct components together.



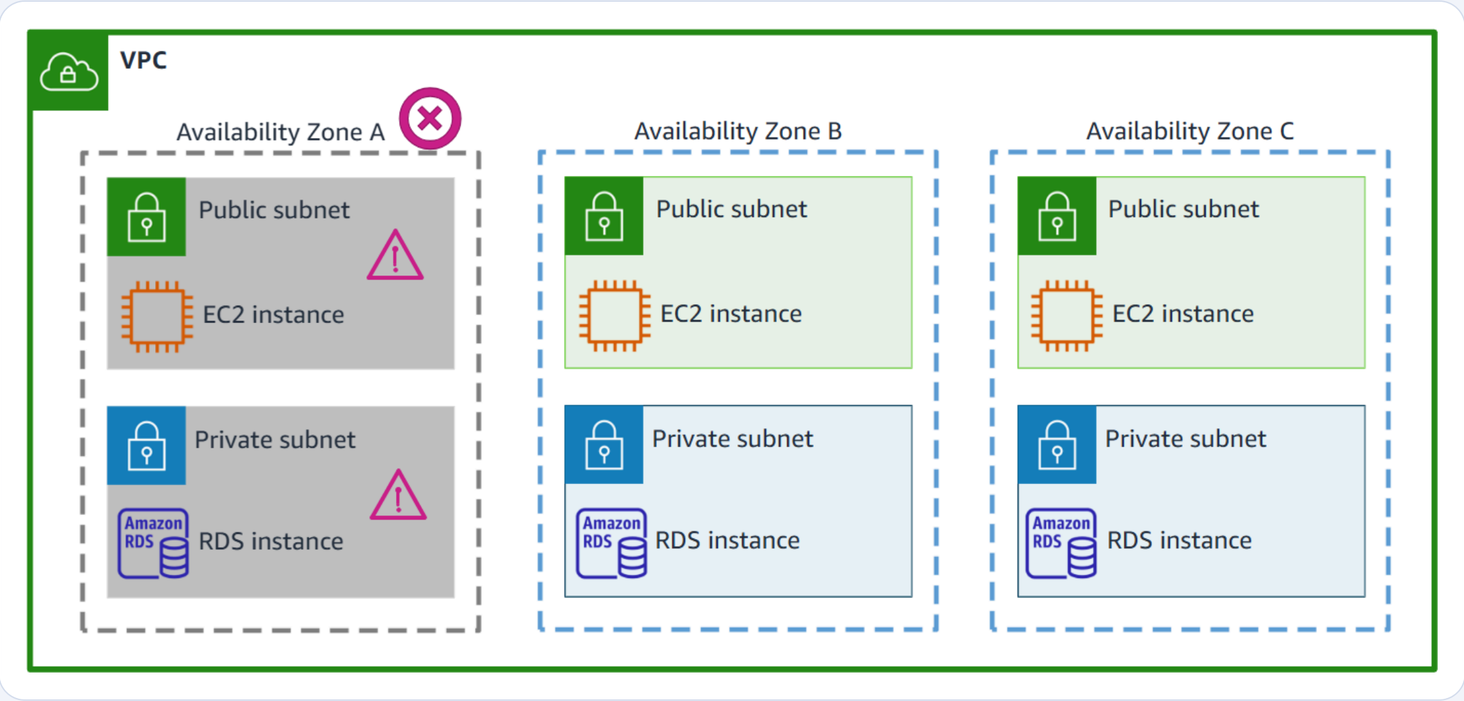
Tagged VPC ID examples	Non-tagged VPC ID examples
vpc-0476e890abedg995f (Development VPC)	vpc-0476e890abedg995f
vpc-s897fsf9afs09h3jk30 (Production VPC)	vpc-s897fsf9afs09h3jk30
vpc-nj789sedjna0halje30 (Testing VPC)	vpc-nj789sedjna0halje30

Subnets in AWS



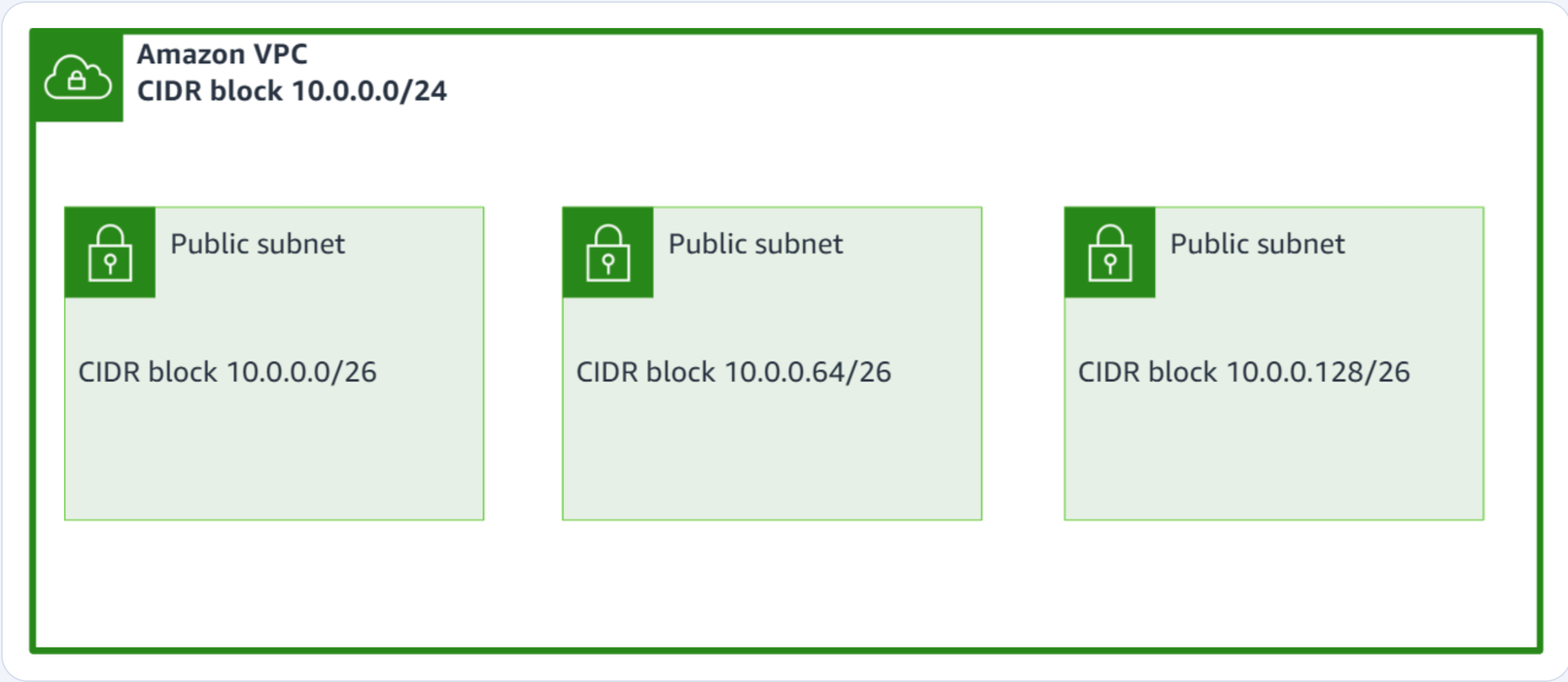
Web servers go in a public subnet. Application and database servers stay in a private subnet.

Subnets in AWS: Availability Zones



Each subnet lives in one Availability Zone. Spread subnets across zones so one failure does not stop the application.

Subnets in AWS: CIDR blocks



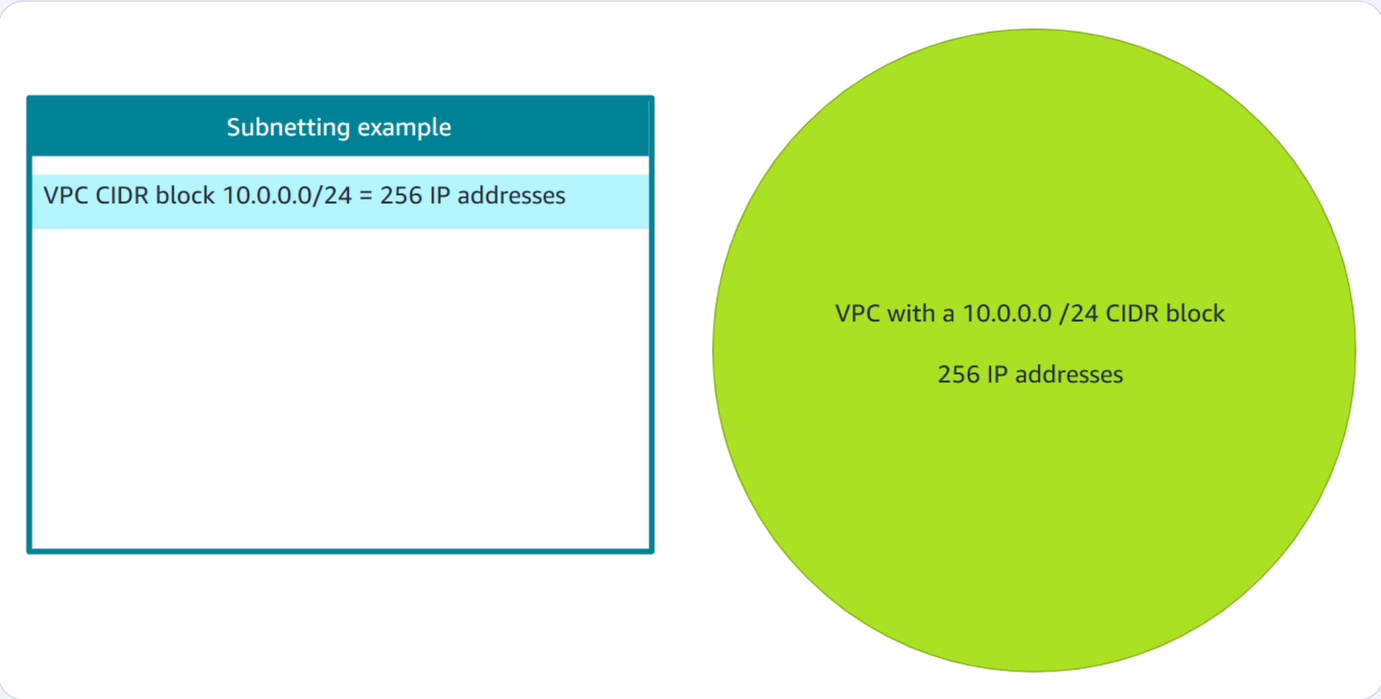
Each subnet’s CIDR block is a part of the VPC’s CIDR block, and the blocks do not overlap.

Reserved subnet IP addresses

Reserved IP address for AWS CIDR blocks	Purpose of reservation
First IP	Network address
Second IP	VPC local router (internal communication)
Third IP	Domain Name System (DNS) resolution
Fourth IP	Future use
Last IP	Network broadcast address

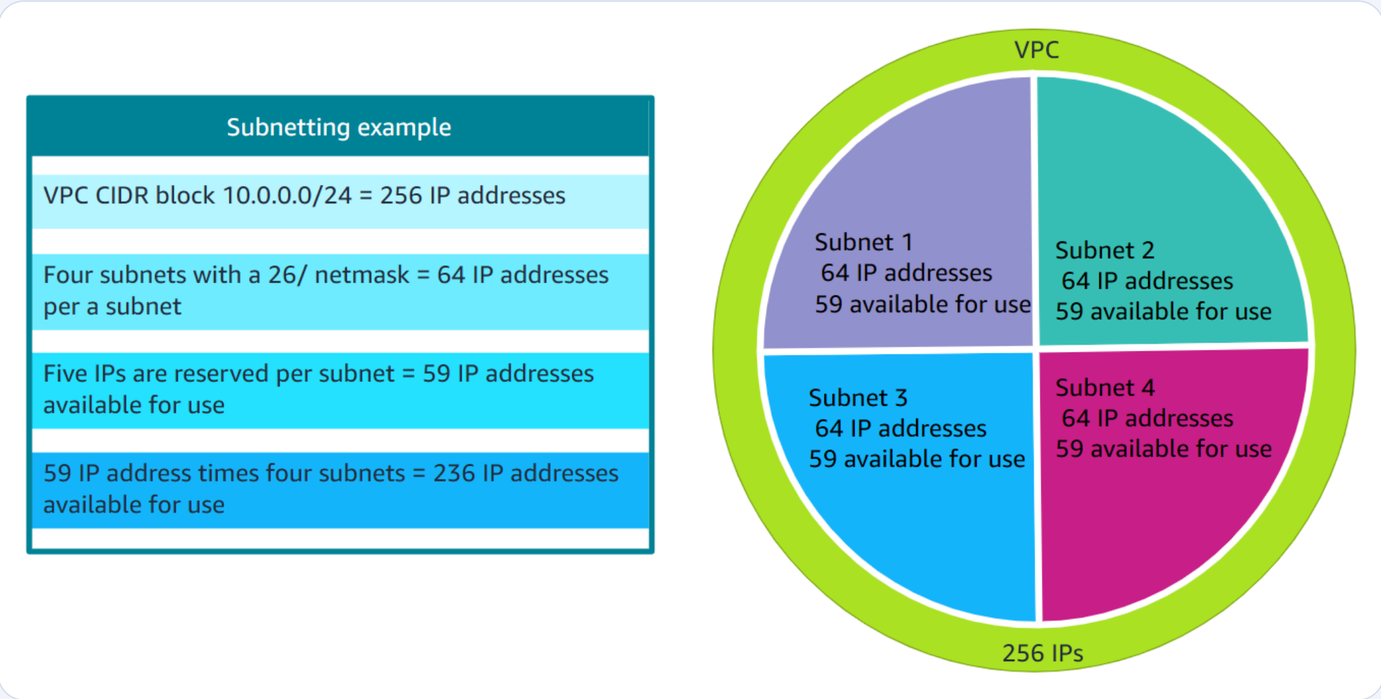
Five IP addresses in every subnet are reserved by AWS, so they are not available for your resources.

Subnetting example



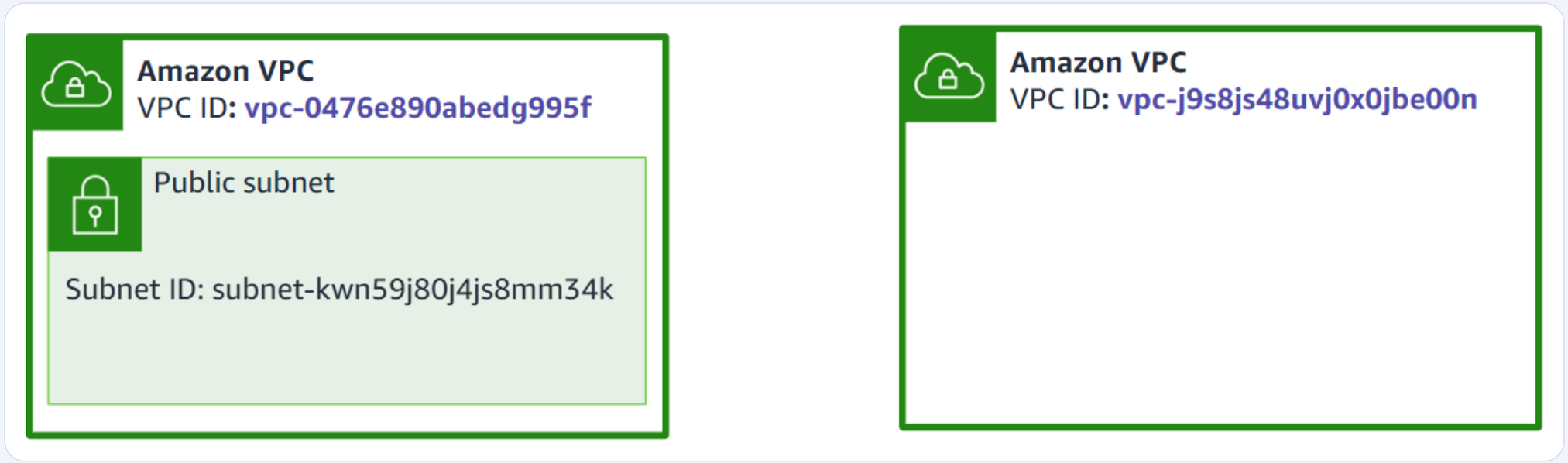
Start with the VPC: CIDR block 10.0.0.0/24 = 256 IP addresses.

Subnetting example: usable addresses



Four /26 subnets of 64 addresses, minus five reserved in each, leave 236 IP addresses available for use.

Subnet and VPC associations

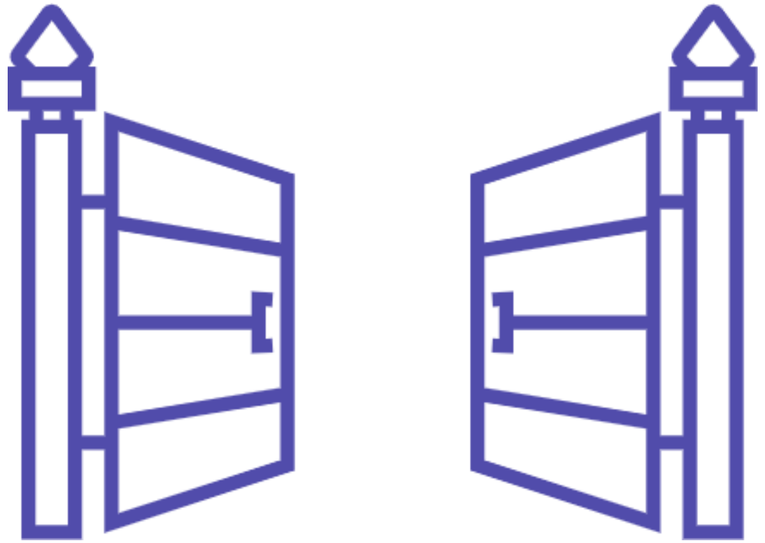


A subnet belongs to exactly one VPC, the one whose VPC ID it was created with.

Network gateways



Internet gateway and virtual private gateway



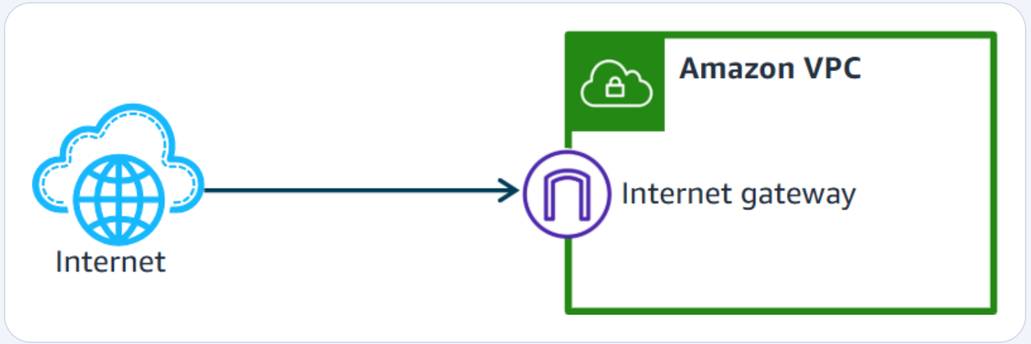
A gateway is the door in and out of the VPC

Internet gateway

An internet gateway is a horizontally scaled, redundant and highly available VPC component that enables communication between your VPC and the internet.

To create one, choose **Create internet gateway**, give it a name and tag, and choose **Create internet gateway**. There are no configuration options to address.

To use it, attach it to your VPC ID. VPCs and internet gateways have a one-to-one relationship. You can detach a gateway and reattach it to another VPC.

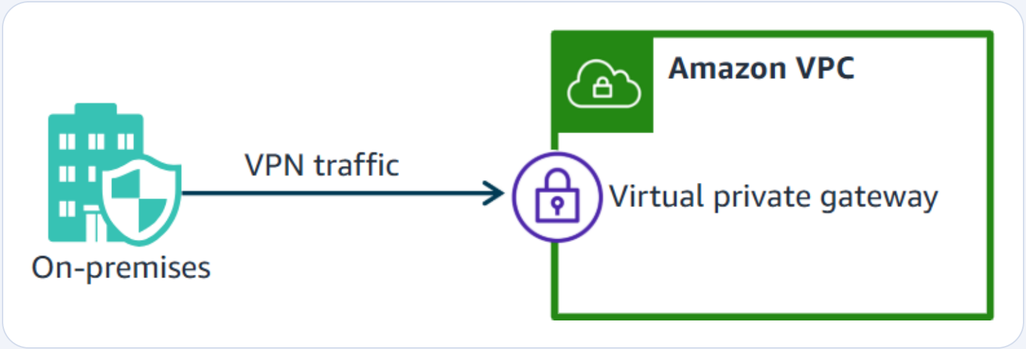


Virtual private gateway

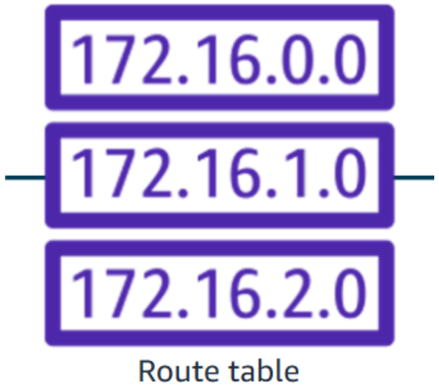
Amazon VPC lets you connect your on-premises site to your VPC. This connection is called a site-to-site VPN connection.

A virtual private gateway is the VPN connector on the Amazon side. You create it and attach it to the VPC ID you want to use for the connection.

For the connection to work, the customer needs a customer gateway on the on-premises side.



Route tables



A route table holds routes. Each route has two parts.

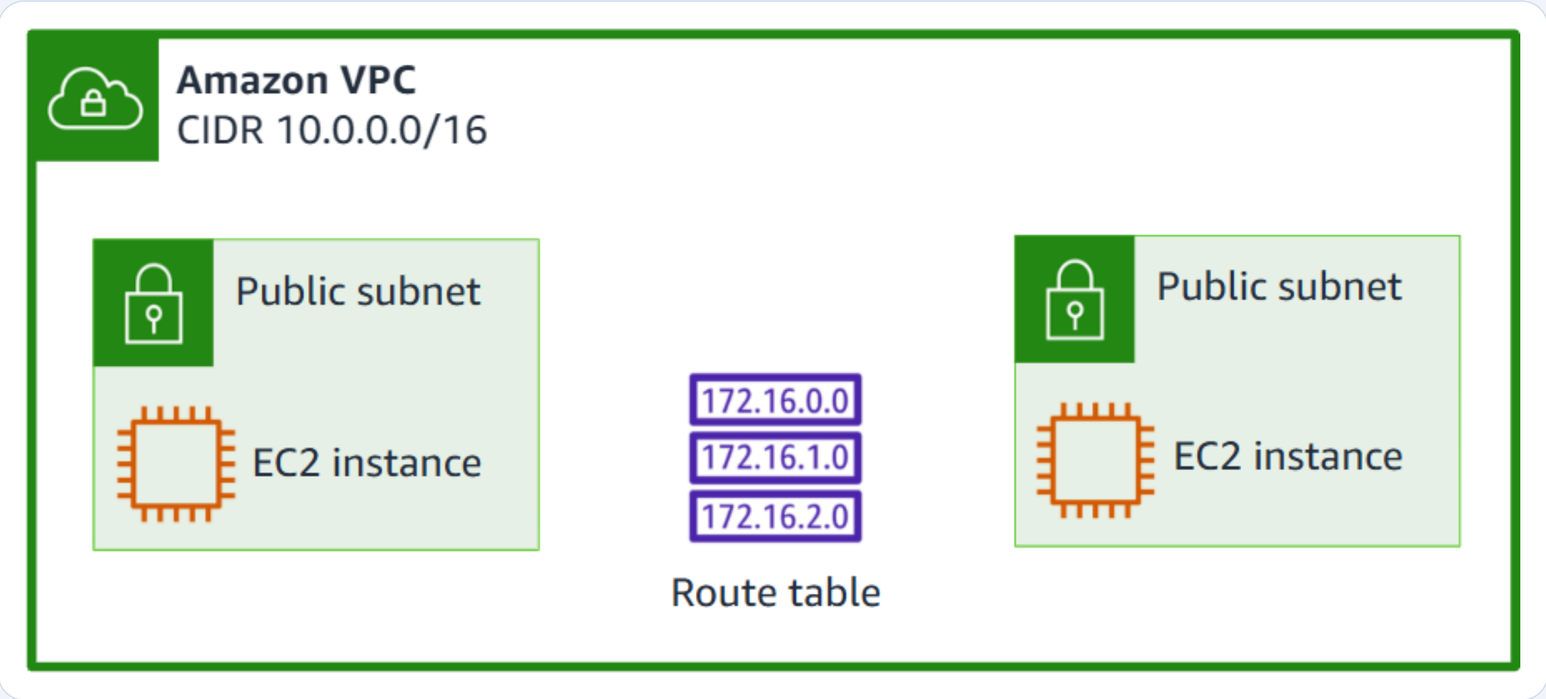
Destination

The range of IP addresses where you want traffic to go (destination CIDR).

Target

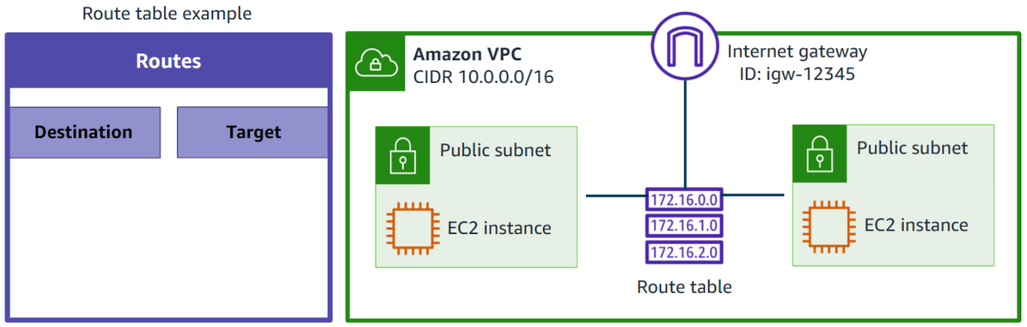
The gateway, network interface or connection through which to send the destination traffic.

Route tables in a VPC

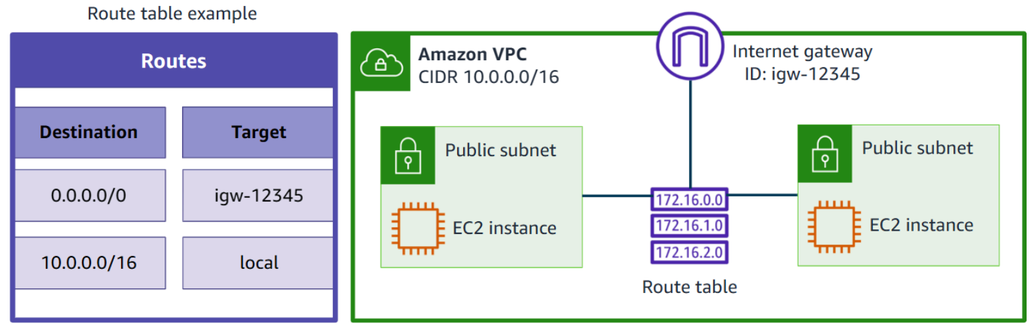


The route table decides where traffic leaving a subnet is sent.

Route table example



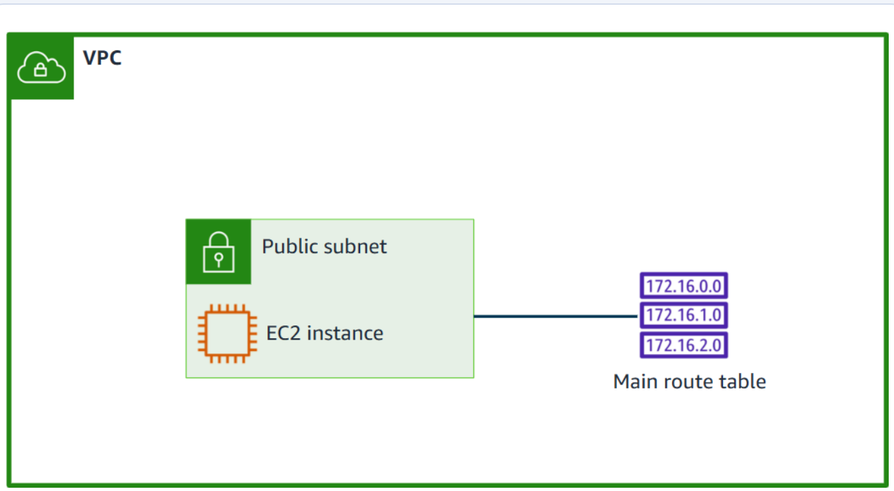
Before: the route table is empty



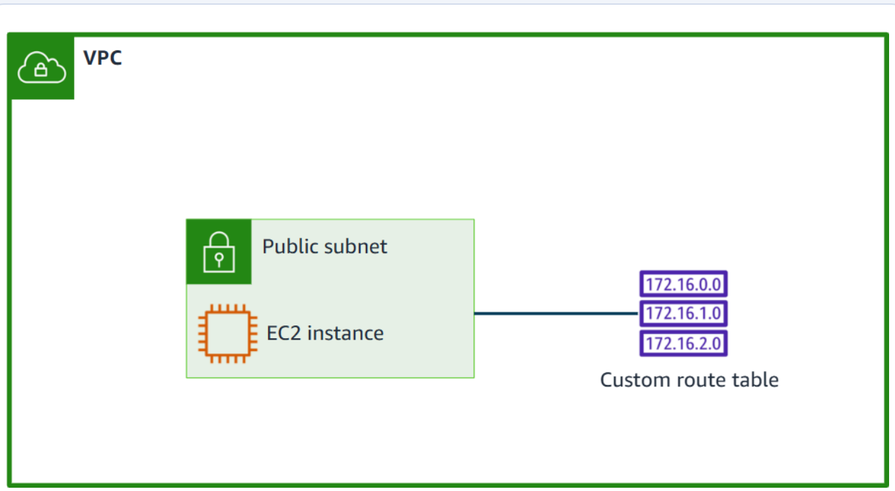
After: two routes

10.0.0.0/16 → local keeps traffic inside the VPC. 0.0.0.0/0 → igw-12345 sends everything else to the internet.

Route table association



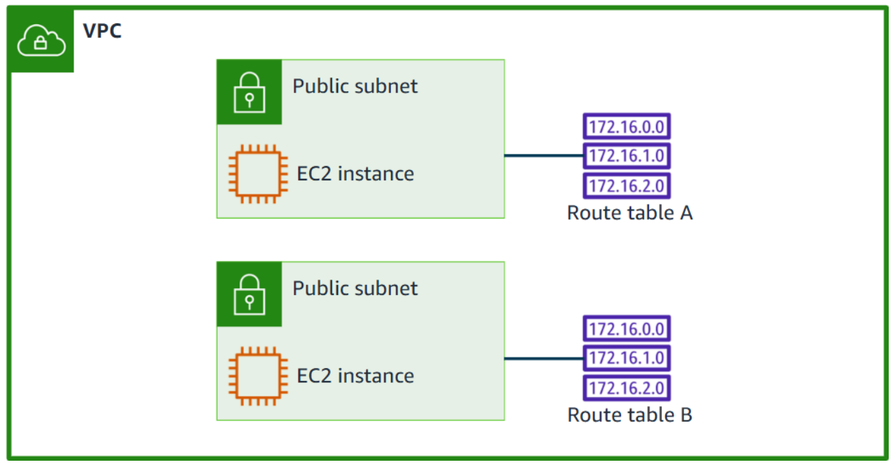
Main route table



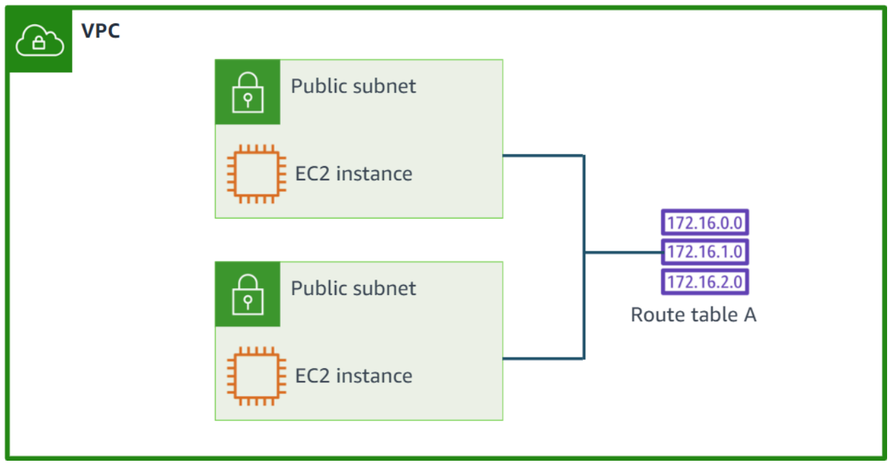
Custom route table

A subnet uses the VPC's main route table unless you associate it with a custom route table.

Route table association: more than one subnet



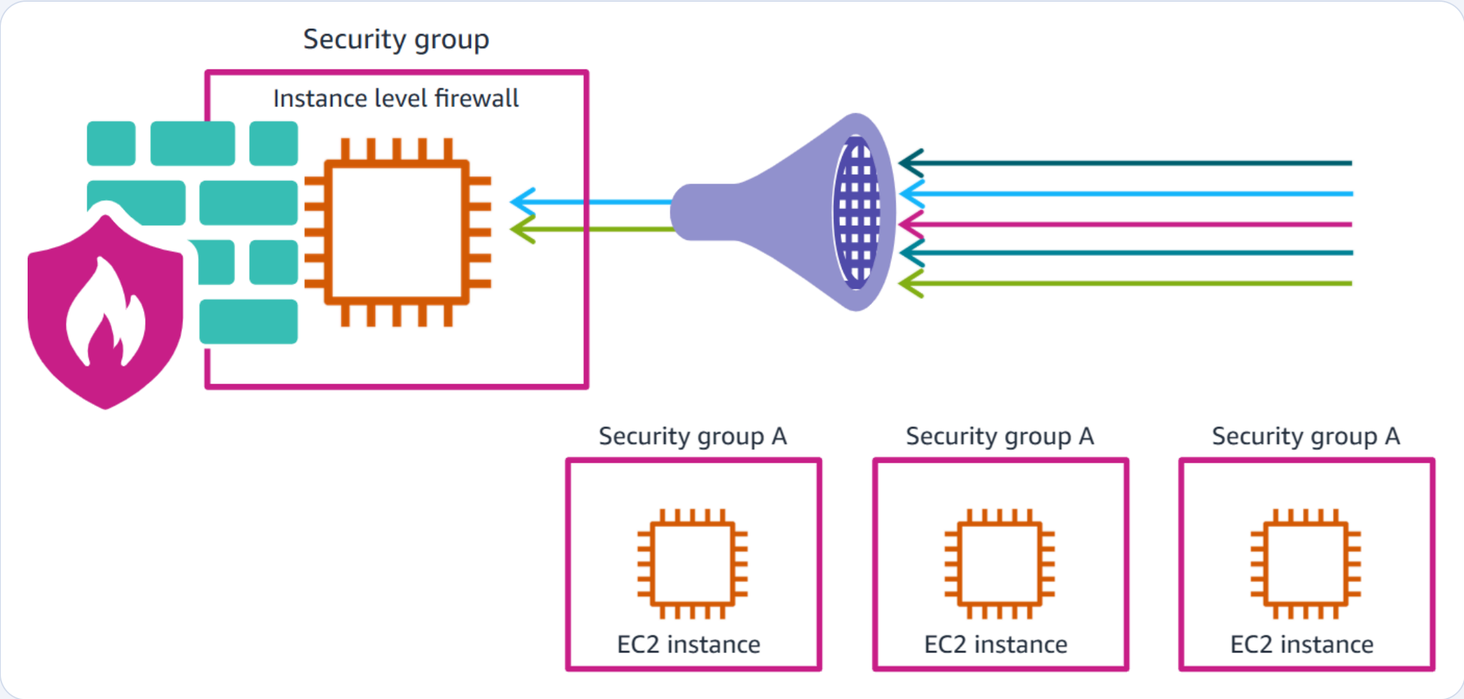
One route table per subnet



Several subnets, one route table

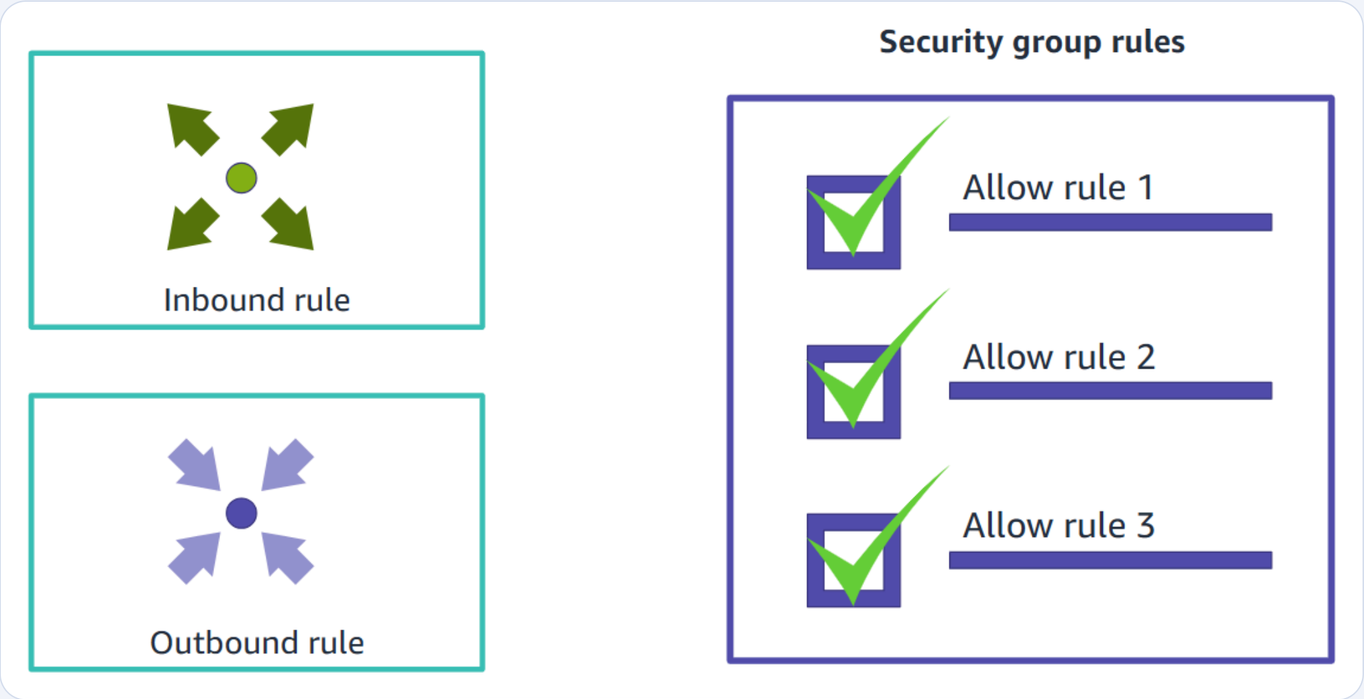
A subnet is associated with one route table at a time. One route table can serve several subnets.

Security groups



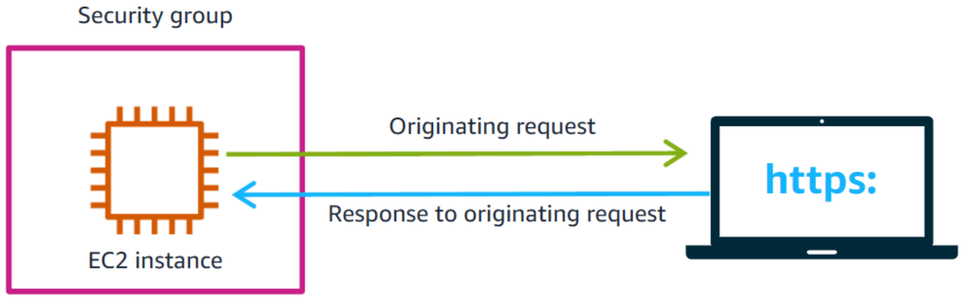
A security group is a firewall at the instance level. Several instances can use the same security group.

Security group rules



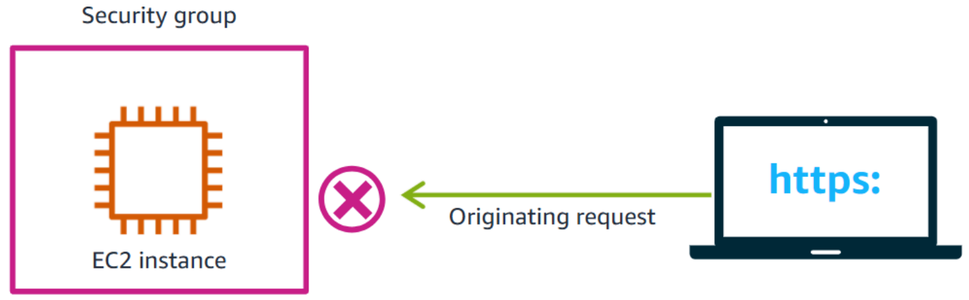
Rules are inbound or outbound, and a security group has allow rules only.

Security groups are stateful



Outbound rule	Inbound rule
Allows all	No rule

The response to our own request is allowed back in

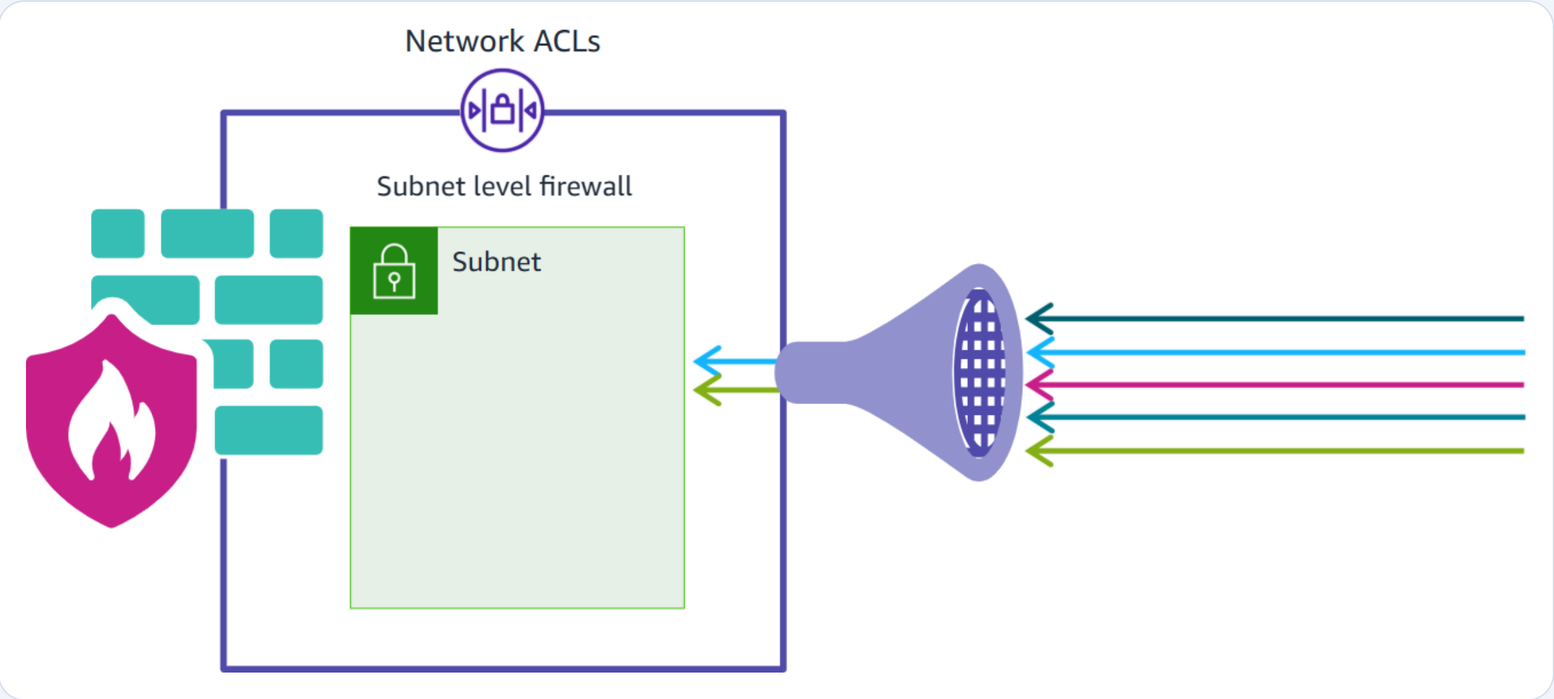


Outbound rule	Inbound rule
Allows all	No rule

A request from outside is blocked

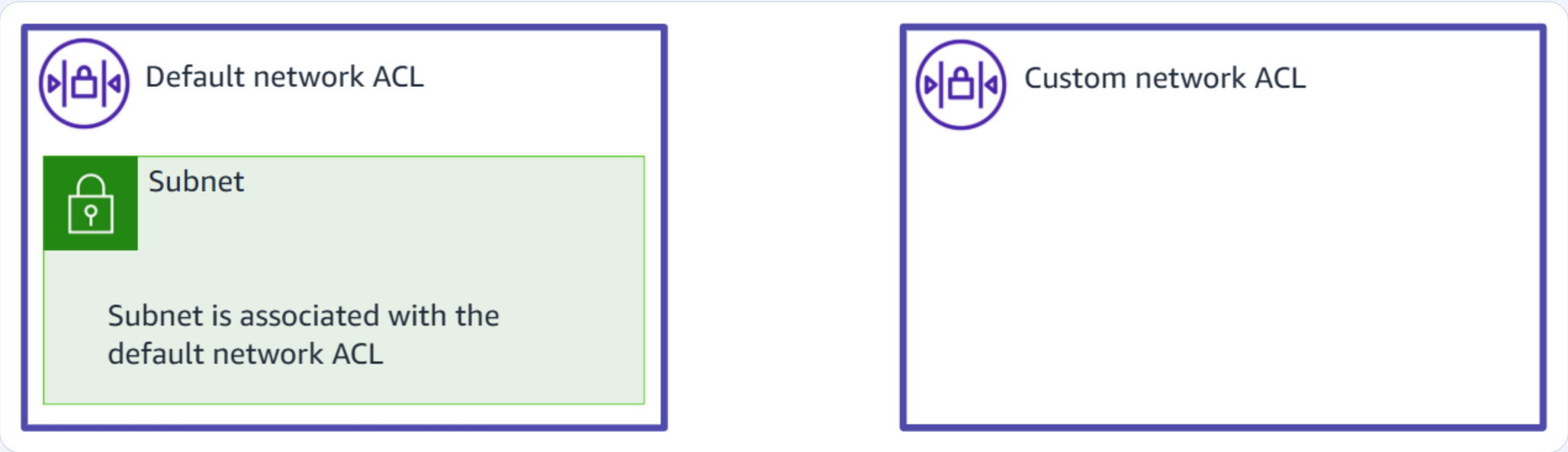
Stateful: return traffic is allowed automatically, regardless of the rules.

Network ACLs



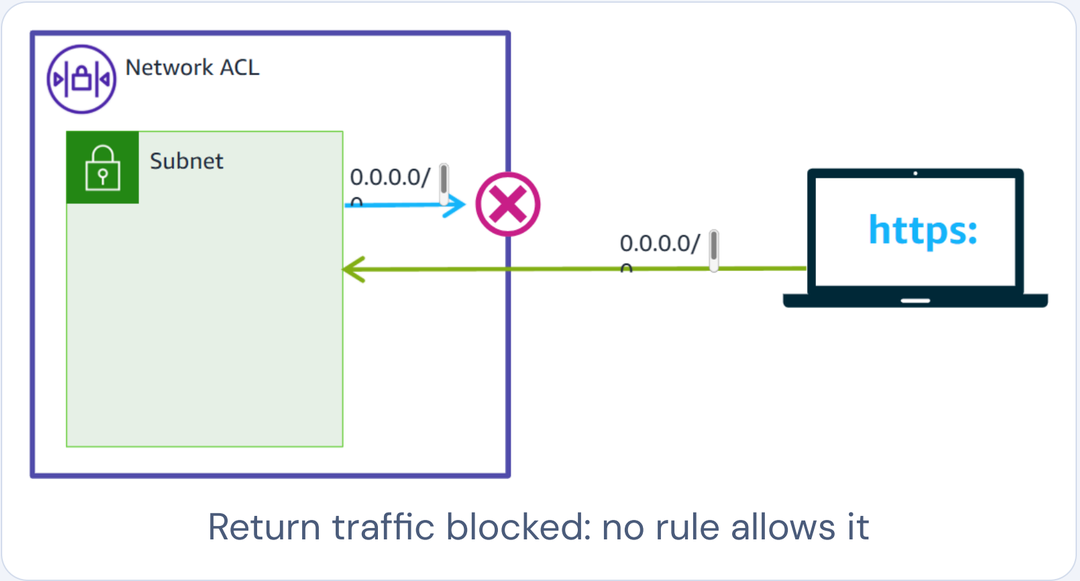
A network access control list (network ACL) is a firewall at the subnet level.

Network ACLs and subnets

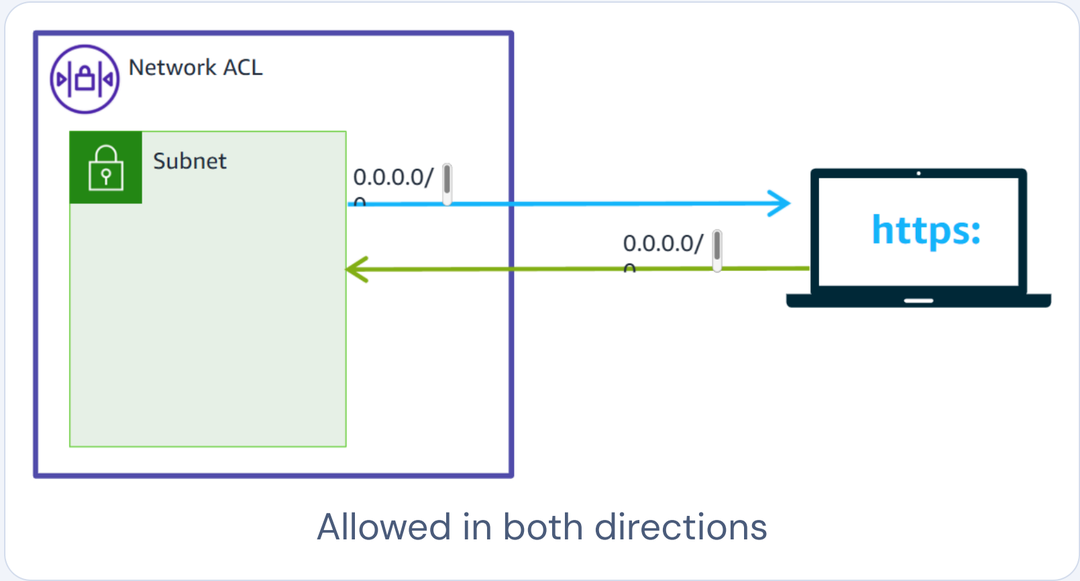


Every subnet is associated with a network ACL. Until you choose another, it is the default network ACL.

Network ACLs are stateless



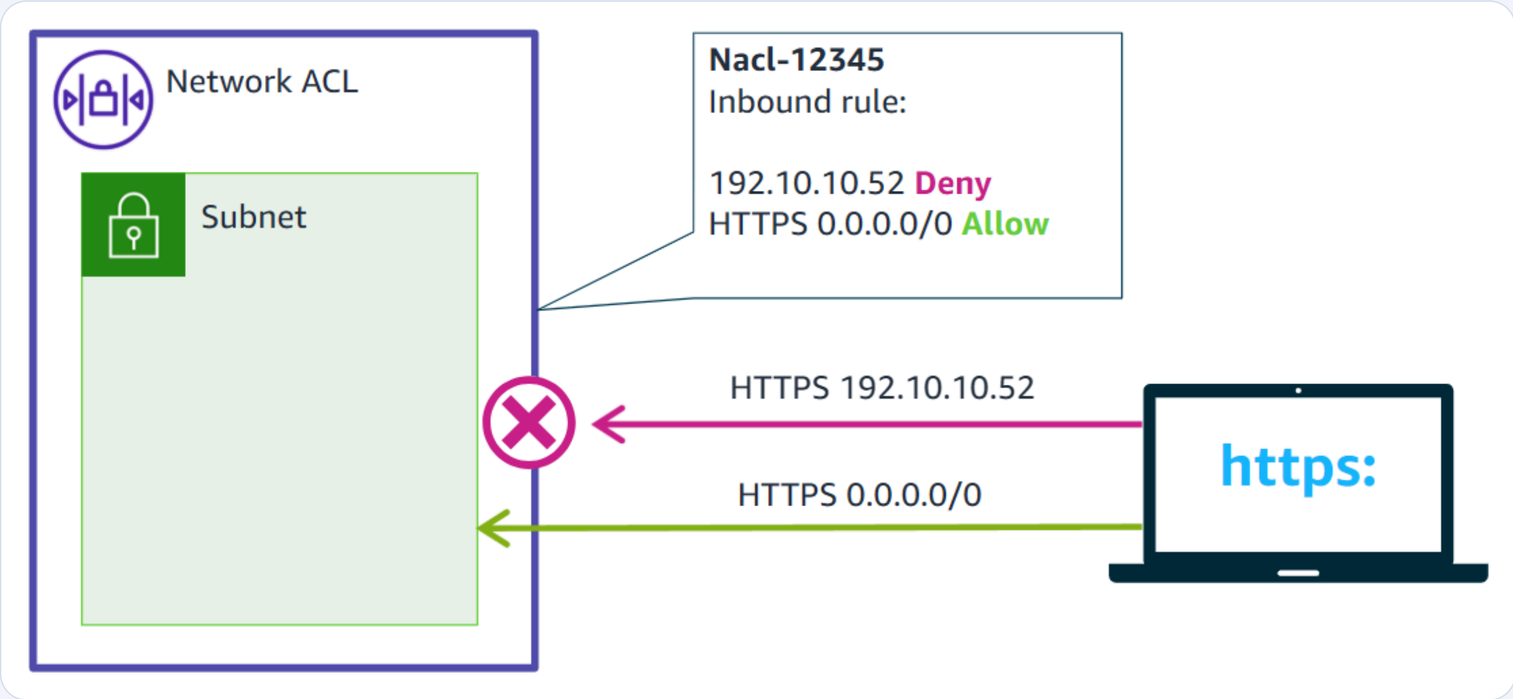
Return traffic blocked: no rule allows it



Allowed in both directions

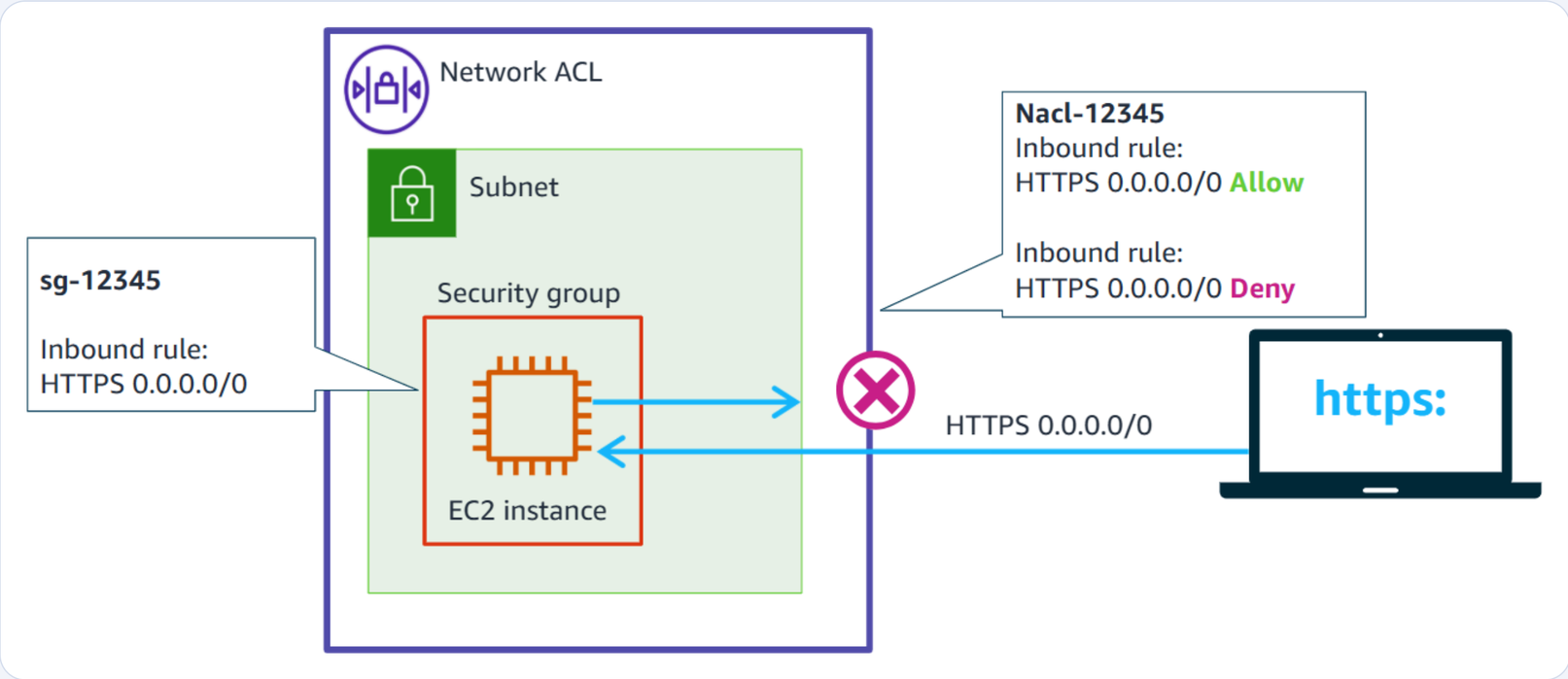
Stateless: return traffic must be explicitly allowed by rules.

Network ACL rules



A network ACL has both allow rules and deny rules, so it can block one specific address.

Network ACL rules with a security group

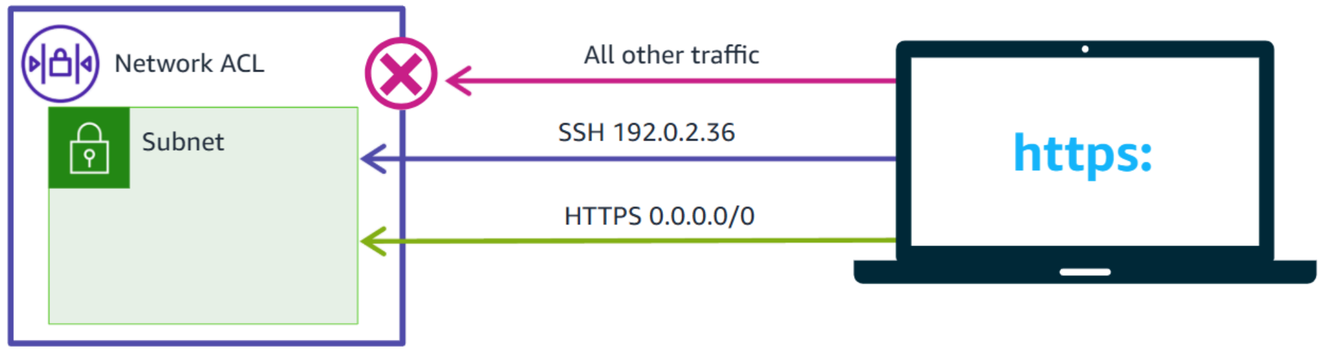


Traffic passes the network ACL at the subnet edge first, then the security group at the instance.

Network ACL rule evaluation: example 1

Inbound rule example 1

Rule	Type	Protocol	Port range	Source	Allow or
10	HTTPS	TCP	443	0.0.0.0/0	Allow
20	SSH	TCP	22	192.0.2.36	Allow
30	All IPV4	All	All	0.0.0.0/0	Deny

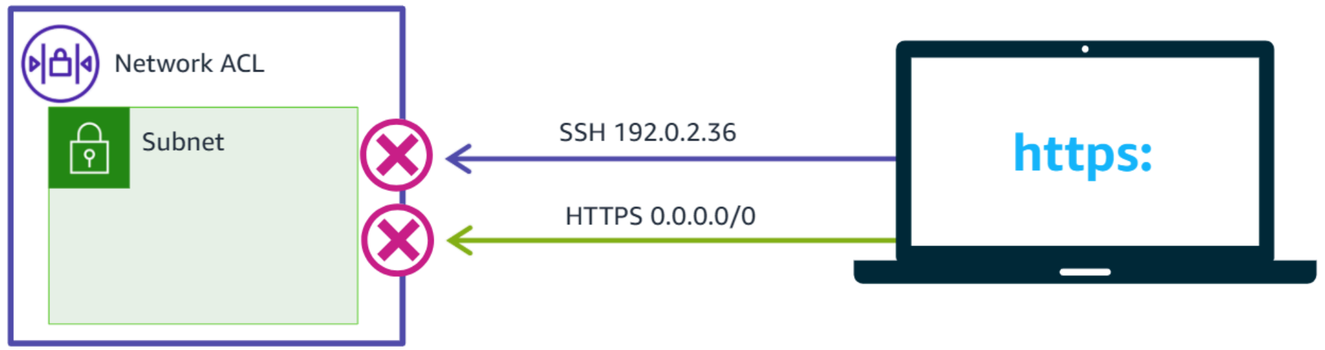


Rules are evaluated in numeric order, lowest first. The first rule that matches decides.

Network ACL rule evaluation: example 2

Inbound rule example 2

Rule	Type	Protocol	Port range	Source	Allow or
10	All IPV4	All	All	0.0.0.0/0	Deny
20	SSH	TCP	22	192.0.2.36	Allow
30	HTTPS	TCP	443	0.0.0.0/0	Allow



Same rules, different order: the deny rule is number 10, so it matches first and everything is blocked.

Default and custom network ACLs

Default network ACL

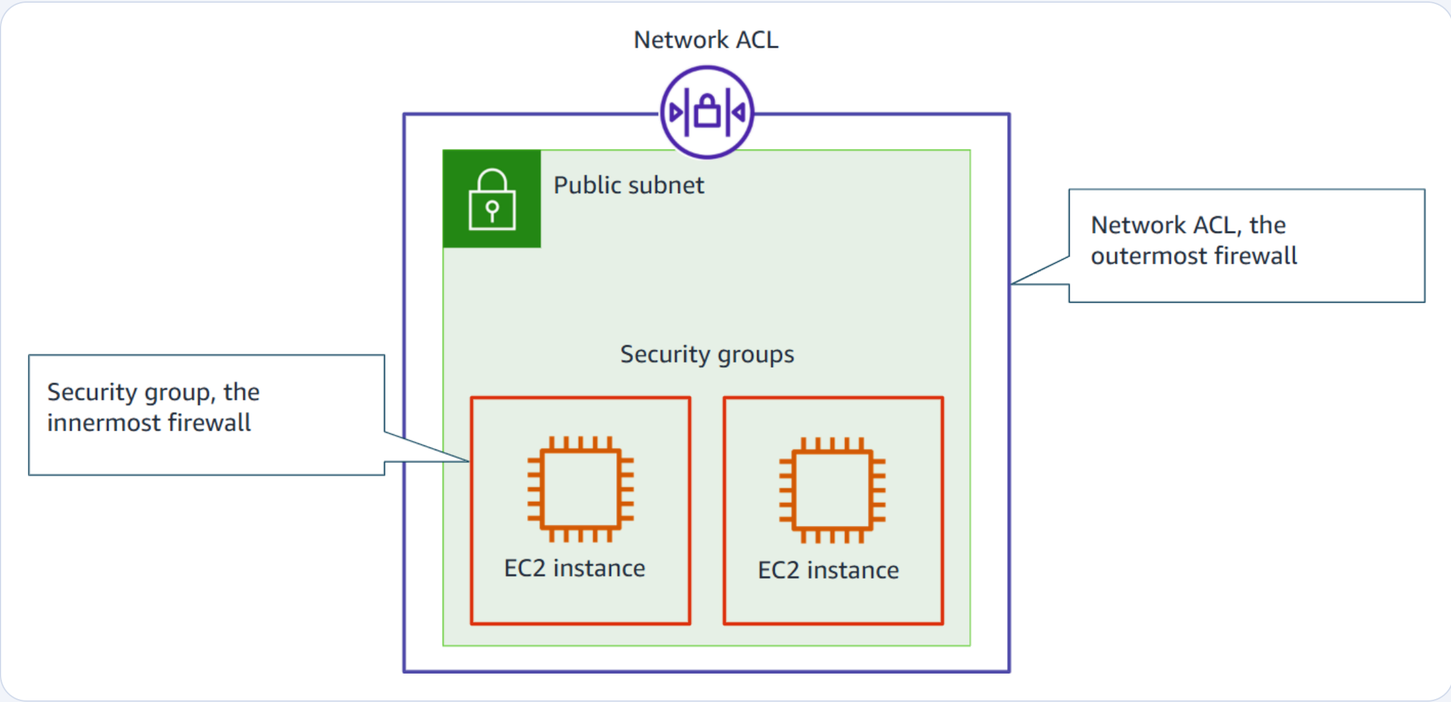
Default settings: allows all inbound and outbound traffic.

Custom network ACL

Default settings: denies all inbound and outbound traffic.

A new custom network ACL blocks everything until you add rules to it.

Security groups vs network ACLs

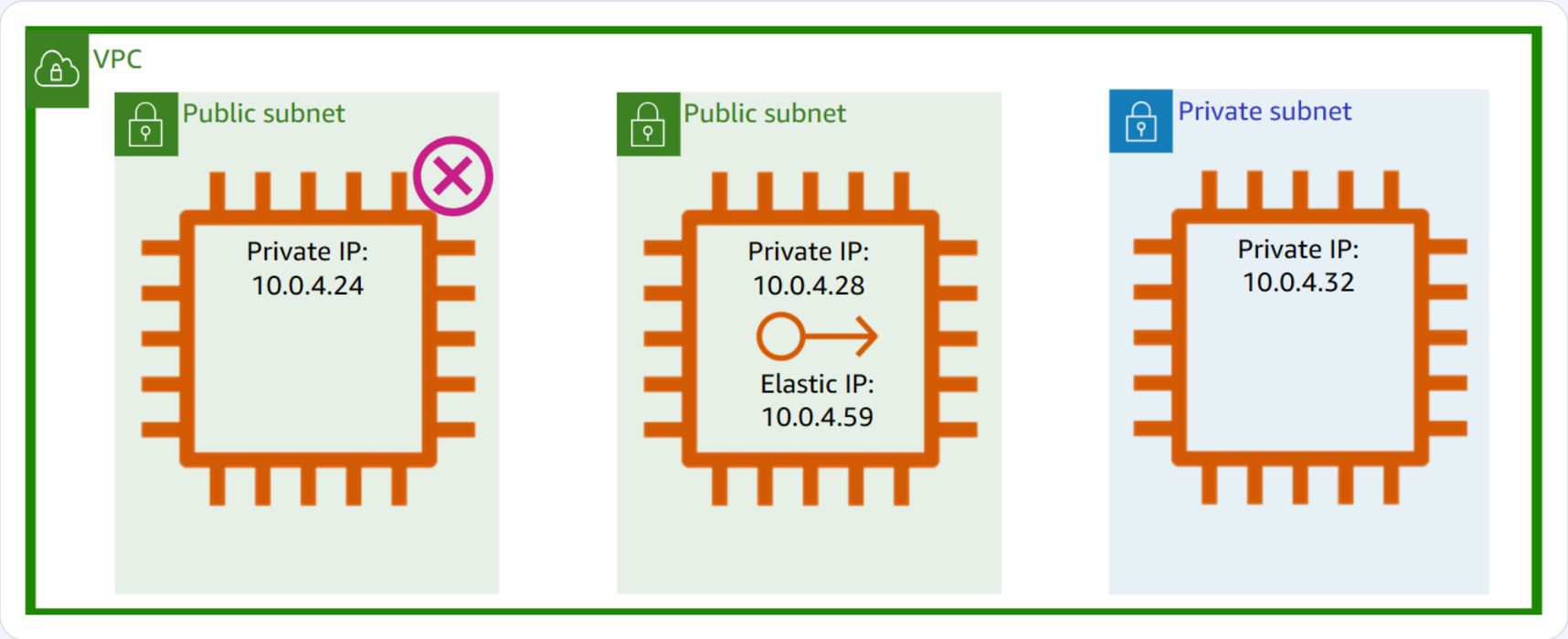


The network ACL is the outermost firewall. The security group is the innermost.

Security groups vs network ACLs

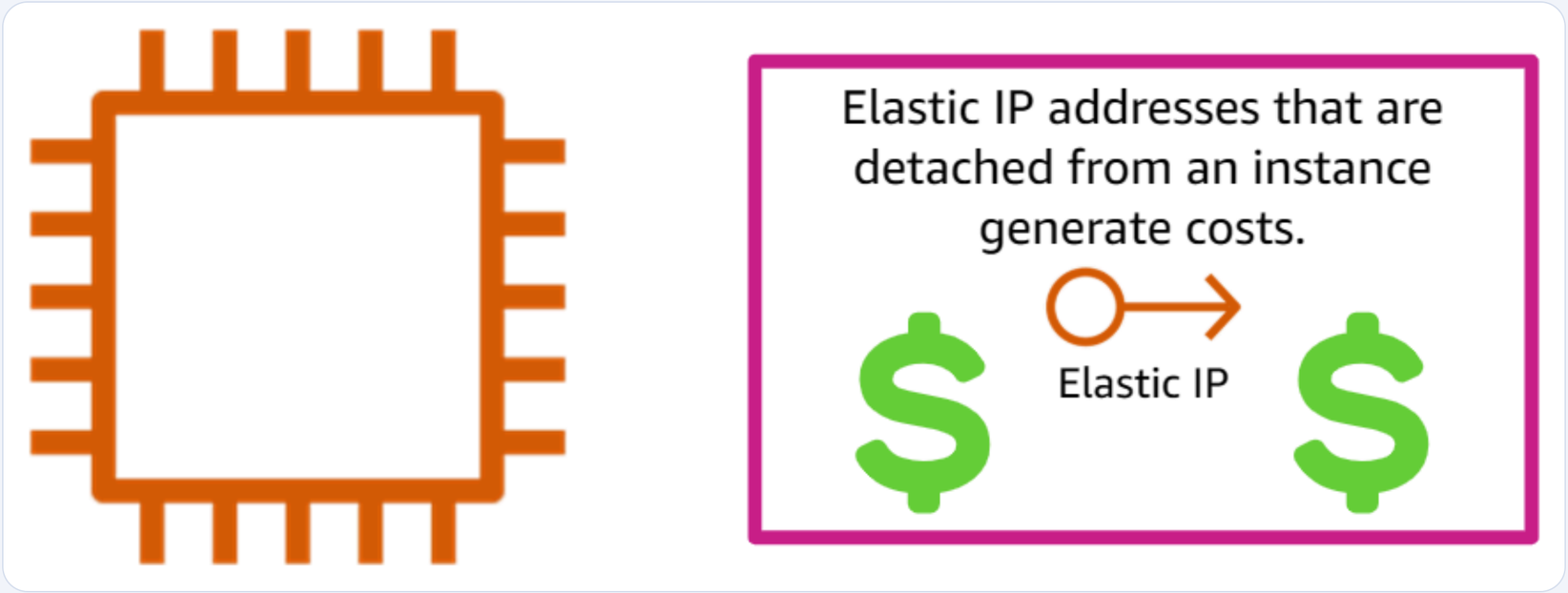
Attribute	Security groups	Network ACLs
Protection level	Instance level (innermost firewall)	Subnet level (outermost firewall)
Usable patterns	Many-to-many: multiple security groups on an instance, and the same group on multiple instances	One-to-many: one network ACL per subnet, and the same ACL on multiple subnets
Flexibility	Can span Availability Zones	Cannot span Availability Zones
Rule types	Allow rules only	Allow and deny rules
State	Stateful: return traffic is allowed automatically	Stateless: return traffic must be explicitly allowed
Decision logic	All rules are evaluated before a decision	Rules are evaluated in numeric order, lowest to highest

Elastic IP address



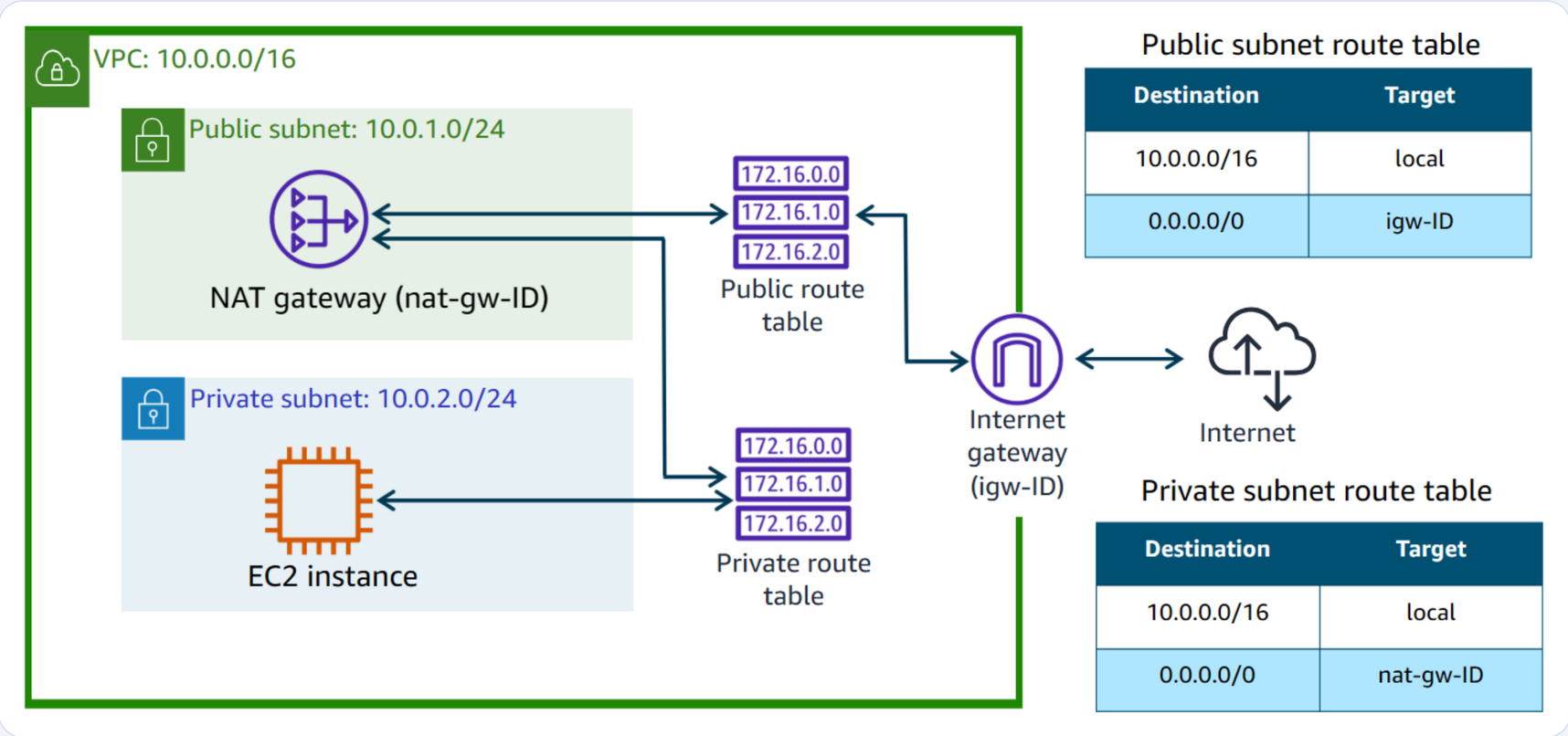
An Elastic IP address is a fixed public address that you associate with an instance.

Elastic IP address: cost

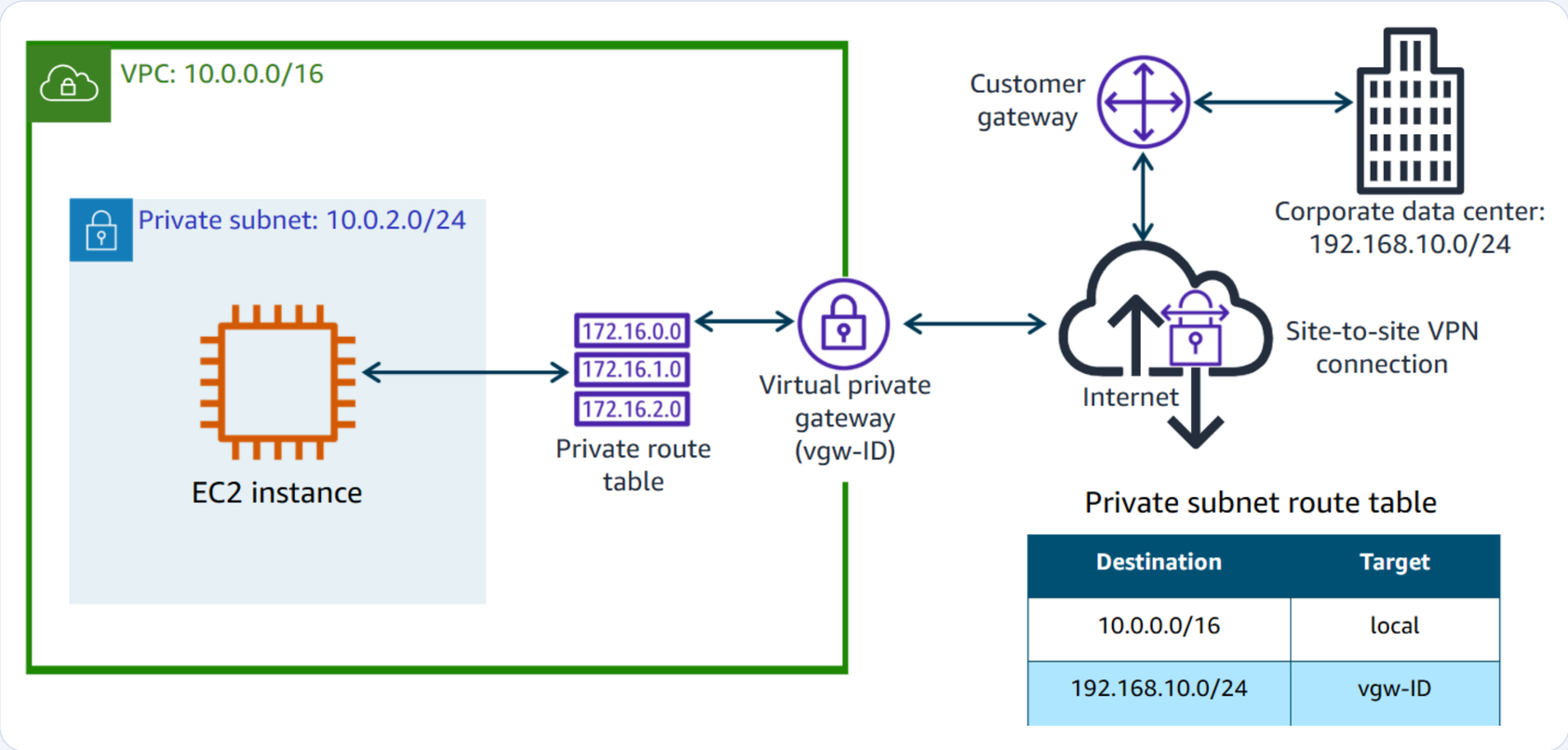


Release an Elastic IP address when it is no longer attached to an instance.

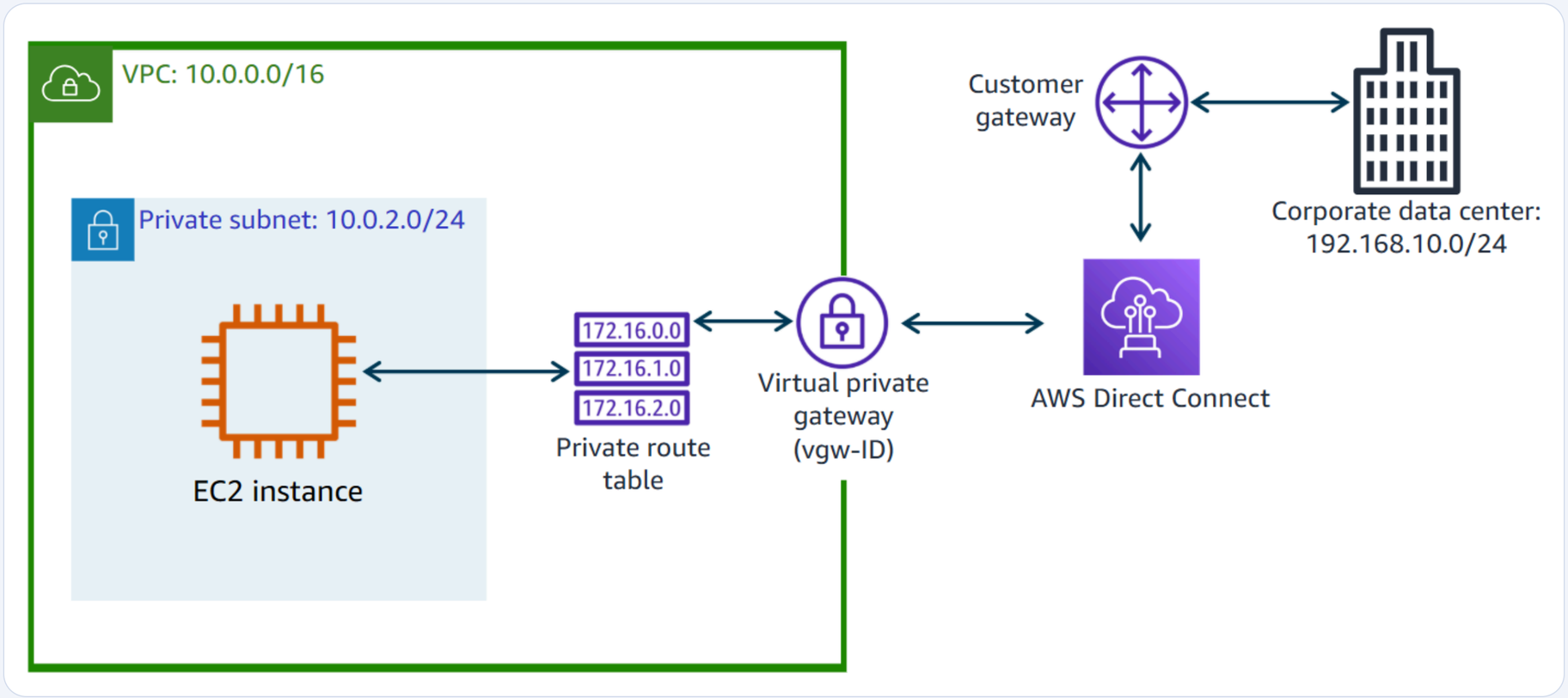
NAT gateway


VPC: 10.0.0.0/16
Public subnet: 10.0.1.0/24
NAT gateway (nat-gw-ID)
Private subnet: 10.0.2.0/24
EC2 instance
172.16.0.0
172.16.1.0
172.16.2.0
Public route table
172.16.0.0
172.16.1.0
172.16.2.0
Private route table
Internet gateway (igw-ID)
Internet
Public subnet route table
Destination
Target
10.0.0.0/16
local
0.0.0.0/0
igw-ID
Private subnet route table
Destination
Target
10.0.0.0/16
local
0.0.0.0/0
nat-gw-ID

AWS VPN


VPC: 10.0.0.0/16
Private subnet: 10.0.2.0/24
EC2 instance
172.16.0.0
172.16.1.0
172.16.2.0
Private route table
Virtual private gateway (vgw-ID)
Customer gateway
Corporate data center: 192.168.10.0/24
Site-to-site VPN connection
Internet
Private subnet route table
Destination
Target
10.0.0.0/16
local
192.168.10.0/24
vgw-ID

AWS Direct Connect



Other topics



Content delivery network

Amazon CloudFront.



Shared responsibility model

QUIZ

Post-knowledge quiz

Networking, CIDR blocks and subnets, and Amazon VPC.

Next session



01

Database